

ISC Class XII Physics

Chapter 9

Ray Optics and Optical Instruments

111 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

- I. Concave Mirror
- II. Convex Mirror
- III. Snell's Law and Refractive Index
- IV. Apparent Depth of Object
- V. Total Internal Reflection
- VI. Refraction at Spherical Surface
- VII. Lens Maker's Formula
- VIII. Effect of Immersing Lens in a Liquid
- IX. Lens Formula
- X. Magnification by the Lens
- XI. Lens Combination (Lenses in Contact)
- XII. Power of the Lens
- XIII. Combination of Lenses and Mirrors
- XIV. Refraction through Prism
- XV. Formula for Refractive Index of Prism
- XVI. Deviation by a Thin Prism
- XVII. Dispersion due to Prism
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Formula Sheet

I–II. Spherical Mirrors

Mirror Formula

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

u = object distance, v = image distance, f = focal length ($f = R/2$), all measured from pole.

Magnification (Mirror)

$$m = -\frac{v}{u} = \frac{f}{f - u}$$

Negative m = real, inverted; positive m = virtual, erect.

III. Snell's Law and Refractive Index

Snell's Law

$$n_1 \sin i = n_2 \sin r$$

i = angle of incidence, r = angle of refraction, n_1, n_2 = refractive indices of the two media.

Refractive Index

$$n = \frac{c}{v} = \frac{\sin i}{\sin r}$$

c = speed of light in vacuum, v = speed in medium.

IV. Apparent Depth

Apparent Depth

$$n = \frac{\text{Real Depth}}{\text{Apparent Depth}}$$

Valid for near-normal viewing from a rarer medium.

Normal Shift

$$\Delta t = t \left(1 - \frac{1}{n} \right)$$

t = thickness of slab/liquid; Δt = apparent upward shift of the object.

V. Total Internal Reflection

Critical Angle

$$\sin C = \frac{1}{n}$$

n = refractive index of denser medium relative to the rarer medium; C = critical angle.

VI. Refraction at a Spherical Surface

Spherical Refraction

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

n_1 = incident-side index, n_2 = refracted-side index, R = radius of curvature of the surface.

VII–VIII. Lens Maker's Formula

Lens Maker's Formula

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

n = refractive index of lens material relative to surrounding medium; R_1, R_2 = radii of the two surfaces.

Lens in a Liquid

$$\frac{1}{f_L} = \left(\frac{n_{\text{lens}}}{n_{\text{liquid}}} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

f_L = focal length of the lens when immersed in a liquid of index n_{liquid} .

IX–XI. Lens Formula, Magnification, Combination

Thin Lens Formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

All distances measured from the optical centre.

Magnification (Lens)

$$m = \frac{v}{u} = \frac{f}{f + u}$$

Positive m = erect (virtual for convex, always virtual for concave); negative m = real, inverted.

Lenses in Contact

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

F = equivalent focal length of two thin lenses held in contact.

XII. Power of a Lens**Power**

$$P = \frac{1}{f \text{ (in metres)}}$$

Unit: dioptre (D). $P_{\text{total}} = P_1 + P_2$ for lenses in contact.

XIII. Combination of Lenses and Mirrors**Sequential Refraction/Reflection**

v_{final} found by applying lens/mirror formula stage by stage

Image of one element becomes the object for the next; effective mirror power $P_{\text{eff}} = P_{\text{lens}} + P_{\text{mirror}} + P_{\text{lens}}$ for lens-mirror silvered systems.

XIV–XVII. Prism**Prism Angle Relation**

$$A = r_1 + r_2$$

A = angle of prism, r_1, r_2 = angles of refraction at the two faces.

Deviation

$$\delta = i + e - A$$

i = angle of incidence, e = angle of emergence.

Refractive Index (Min. Deviation)

$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

δ_m = angle of minimum deviation (occurs when $i = e$, $r_1 = r_2$).

Thin Prism Deviation

$$\delta = (n - 1)A$$

Valid for small prism angle A (thin prism approximation).

Angular Dispersion

$$\theta = (n_v - n_r)A$$

n_v, n_r = refractive indices for violet and red light.

XVIII–XIX. Microscopes**Simple Microscope**

$$M = 1 + \frac{D}{f} \text{ (image at } D\text{)}; \quad M = \frac{D}{f} \text{ (image at } \infty\text{)}$$

D = least distance of distinct vision (25 cm), f = focal length of lens.

Compound Microscope

$$M = m_o \times m_e = \frac{v_o}{u_o} \times \left(1 + \frac{D}{f_e}\right)$$

m_o = linear magnification of objective, m_e = angular magnification of eyepiece.

XX. Telescope**Astronomical Telescope**

$$M = -\frac{f_o}{f_e} \text{ (relaxed eye)}; \quad L = f_o + f_e$$

f_o, f_e = focal lengths of objective and eyepiece; L = tube length for image at infinity.

Telescope (Image at D)

$$M = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{D}\right)$$

Used when the final image is formed at the least distance of distinct vision.

I. Concave Mirror

Question 1

An object is placed in front of a concave mirror of radius of curvature 40 cm at a distance of 10 cm. Find the position, nature and magnification of the image.

Given: $R = 40 \text{ cm} \Rightarrow f = -20 \text{ cm}$ (concave); $u = -10 \text{ cm}$

Formula: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}, \quad m = -\frac{v}{u}$

Solution:

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-20} - \frac{1}{-10} = -\frac{1}{20} + \frac{1}{10} = \frac{1}{20}$$

$\Rightarrow v = +20 \text{ cm}$ (positive $\Rightarrow$ behind the mirror, virtual)

$$m = -\frac{v}{u} = -\frac{20}{-10} = +2$$

Final Answer

Image is 20 cm behind the mirror, virtual, erect, magnification = +2.

Common Mistake: Do not forget the sign of f : for a concave mirror f is always negative in the Cartesian convention.

Question 2

A 1 cm long flame is placed at a distance of 1.5 m from a wall. How far from the wall should a concave mirror be placed in order to obtain a 2 cm long flame on the same wall? What is the focal length of the mirror?

Given: Flame-wall distance = 1.5 m; image (on wall) is real, $|m| = 2$

Formula: $|u| + |v| = 1.5 \text{ m}, \quad |m| = \frac{|v|}{|u|} = 2$

Solution:

Let mirror be at distance d from the wall. Then $|v| = d$ and $|u| = 1.5 - d$.

Since $|v| = 2|u|$: $d = 2(1.5 - d) \Rightarrow 3d = 3 \Rightarrow d = 1 \text{ m}$

So $|u| = 0.5 \text{ m}$, $|v| = 1 \text{ m}$ (both real, so $u = -0.5 \text{ m}$, $v = -1 \text{ m}$)

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} = \frac{1}{-1} + \frac{1}{-0.5} = -1 - 2 = -3 \Rightarrow f = -\frac{1}{3} \text{ m}$$

Final Answer

Mirror should be placed 1 m from the wall; $f = \frac{1}{3} \text{ m} \approx 33.3 \text{ cm}$ (both real/negative).

Question 3

An object is placed at a distance of 30 cm from a concave mirror of focal length 15 cm. If the object is displaced through a distance of 10 cm away from the mirror, find the displacement of the image.

★ Likely Exam Question

Given: $f = -15 \text{ cm}$; $u_1 = -30 \text{ cm}$; $u_2 = -40 \text{ cm}$

Formula: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

Solution:

Position 1: $\frac{1}{v_1} = \frac{1}{-15} - \frac{1}{-30} = -\frac{1}{15} + \frac{1}{30} = -\frac{1}{30} \Rightarrow v_1 = -30 \text{ cm}$

Position 2: $\frac{1}{v_2} = \frac{1}{-15} - \frac{1}{-40} = -\frac{1}{15} + \frac{1}{40} = -\frac{1}{24} \Rightarrow v_2 = -24 \text{ cm}$

Displacement of image = $|v_1 - v_2| = |-30 - (-24)| = 6 \text{ cm}$; since $|v|$ decreased, the image has moved towards the mirror.

Final Answer

Image displacement = 6 cm towards the mirror.

Common Mistake: Students often assume the image moves the same direction and distance as the object — this is only true far from f and $2f$; always recompute v separately.

Question 4

A concave mirror has a radius of curvature of 30 cm. How far must it be placed from a small object in order to give a five times magnified virtual image?

Given: $R = 30 \text{ cm} \Rightarrow f = -15 \text{ cm}$; $m = +5$ (virtual, erect)

Formula: $m = \frac{f}{f - u}$

Solution:

$$5 = \frac{-15}{-15 - u} \Rightarrow -15 - u = \frac{-15}{5} = -3 \Rightarrow -u = -3 + 15 = 12 \Rightarrow u = -12 \text{ cm}$$

Final Answer

Object should be placed 12 cm in front of the mirror.

Question 5

A concave mirror produces a real image of half the size of an object placed at 60 cm in front of it. Where should the object be placed to obtain a virtual image of double the size of the object?

Given: $u_1 = -60 \text{ cm}$, $m_1 = -\frac{1}{2}$ (real); required $m_2 = +2$ (virtual)

Formula: $m = -\frac{v}{u}$, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

Solution:

Step 1 — find f : $-\frac{1}{2} = -\frac{v_1}{-60} \Rightarrow v_1 = -30 \text{ cm}$

$$\frac{1}{f} = \frac{1}{-30} + \frac{1}{-60} = -\frac{1}{20} \Rightarrow f = -20 \text{ cm}$$

Step 2 — for virtual image, $m_2 = -\frac{v_2}{u_2} = 2 \Rightarrow v_2 = -2u_2$. Substituting in mirror formula:

$$\frac{1}{-2u_2} + \frac{1}{u_2} = \frac{1}{f} \Rightarrow \frac{1}{2u_2} = \frac{1}{f} \Rightarrow u_2 = \frac{f}{2} = \frac{-20}{2} = -10 \text{ cm}$$

Final Answer

Object should be placed 10 cm in front of the mirror.

Common Mistake: A virtual, magnified image in a concave mirror is only possible when the object lies between the pole and the focus — always check $|u| < |f|$.

Question 6

An object is placed first at a distance of 25 cm and then at a distance of 40 cm from a concave mirror. The magnification in the first case is four times than that in the second case, the image being real in both cases. What is the focal length of the mirror?

★ Likely Exam Question

Given: $u_1 = -25 \text{ cm}$, $u_2 = -40 \text{ cm}$, $m_1 = 4m_2$ (both real)

Formula: $m = \frac{f}{f - u}$

Solution:

$$\frac{f}{f + 25} = 4 \times \frac{f}{f + 40}$$

$$\frac{1}{f + 25} = \frac{4}{f + 40}$$

$$f + 40 = 4(f + 25) = 4f + 100$$

$$-60 = 3f \Rightarrow f = -20 \text{ cm}$$

Final Answer

Focal length of the mirror = -20 cm .

Question 7

A concave mirror for face viewing has focal length 36.0 cm . Find the distance at which you should hold the mirror from your face in order to see your image upright with a magnification of 4.

Given: $f = -36 \text{ cm}$; $m = +4$ (virtual, erect)

Formula: $m = \frac{f}{f - u}$

Solution:

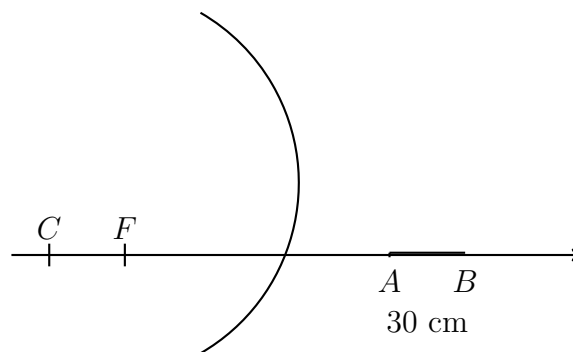
$$4 = \frac{-36}{-36 - u} \Rightarrow -36 - u = \frac{-36}{4} = -9 \Rightarrow u = -36 + 9 = -27 \text{ cm}$$

Final Answer

Mirror should be held 27.0 cm from the face.

Question 8

An object AB is placed along the axis of a concave mirror of focal length 10 cm . End A (nearer end) of the object is at 30 cm from the mirror. Find the length of the image, if: (i) length of object is 5 cm , (ii) length of object is 1 mm , and (iii) show the image's position in case (i).



Object AB (5 cm) placed along the principal axis, near end A at 30 cm from a concave mirror of $f = 10$ cm

Given: $f = -10$ cm; $u_A = -30$ cm; object length (i) 5 cm, (ii) 1 mm

Formula: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$, longitudinal magnification $m_L = \left(\frac{v}{u}\right)^2$

Solution:

(i) End A : $u = -30$: $\frac{1}{v_A} = \frac{1}{-10} - \frac{1}{-30} = -\frac{1}{15} \Rightarrow v_A = -15$ cm

End B (far end, $u = -35$): $\frac{1}{v_B} = \frac{1}{-10} - \frac{1}{-35} = -\frac{1}{14} \Rightarrow v_B = -14$ cm

Image length $= |v_A - v_B| = |-15 - (-14)| = 1$ cm

(ii) Near A : $m_L = \left(\frac{v_A}{u_A}\right)^2 = \left(\frac{-15}{-30}\right)^2 = 0.25$

$dv = m_L \times du = 0.25 \times 1 \text{ mm} = 0.25 \text{ mm}$

Final Answer

(i) Image length = 1 cm (along the principal axis); (ii) Image length = 0.25 mm; (iii) Image $A'B'$ is real, inverted, and lies between 14 cm and 15 cm from the mirror.

Common Mistake: Longitudinal magnification is $(v/u)^2$, NOT the same as lateral magnification $-v/u$ — a very common exam slip.

II. Convex Mirror

Question 9

An object is placed at 18 cm in front of a spherical mirror. The image is formed at 3.0 cm behind the mirror. Find the focal length and the nature of the mirror.

Given: $u = -18$ cm; $v = +3$ cm (behind mirror $\Rightarrow$ virtual)

Formula: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

Solution:

$$\frac{1}{f} = \frac{1}{3} + \frac{1}{-18} = \frac{6}{18} - \frac{1}{18} = \frac{5}{18}$$

$$f = \frac{18}{5} = 3.6 \text{ cm (positive } \Rightarrow \text{ convex mirror)}$$

Final Answer

Focal length = +3.6 cm; mirror is convex.

Question 10

An object is placed 15 cm away from a convex mirror of radius of curvature 90 cm. Find the image's position and magnification.

Given: $R = 90$ cm $\Rightarrow f = +45$ cm; $u = -15$ cm

Formula: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}, \quad m = -\frac{v}{u}$

Solution:

$$\frac{1}{v} = \frac{1}{45} - \frac{1}{-15} = \frac{1}{45} + \frac{3}{45} = \frac{4}{45}$$

$$v = \frac{45}{4} = 11.25 \text{ cm (behind the mirror, virtual)}$$

$$m = -\frac{11.25}{-15} = +0.75$$

Final Answer

Image is 11.25 cm behind the mirror, virtual; magnification = +0.75.

Question 11

A 2 cm long object is placed at a distance of 100 cm from a convex mirror of radius of curvature 50 cm. Find the position, nature and length of the image.

Given: $R = 50 \text{ cm} \Rightarrow f = +25 \text{ cm}$; $u = -100 \text{ cm}$; object height = 2 cm

Formula: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}, \quad m = -\frac{v}{u}$

Solution:

$$\frac{1}{v} = \frac{1}{25} - \frac{1}{-100} = \frac{4}{100} + \frac{1}{100} = \frac{5}{100}$$

$v = 20 \text{ cm}$ (behind mirror, virtual)

$$m = -\frac{20}{-100} = +0.2$$

Image length = $2 \times 0.2 = 0.4 \text{ cm}$

Final Answer

Image is 20 cm behind the mirror, virtual, erect, length = 0.4 cm.

III. Snell's Law and Refractive Index

Question 12

The refractive index of diamond relative to air is 2.42 and that of glass is 1.51. How much faster does light travel in glass than in diamond?

Given: $n_d = 2.42$; $n_g = 1.51$

Formula: $n = \frac{c}{v} \Rightarrow v = \frac{c}{n}$

Solution:

$$\frac{v_g}{v_d} = \frac{c/n_g}{c/n_d} = \frac{n_d}{n_g} = \frac{2.42}{1.51} = 1.60$$

Final Answer

Light travels 1.60 times faster in glass than in diamond.

Question 13

A ray of light in air is incident on a glass plate at 45° . The refractive index of glass is 1.5. What is the angle of refraction?

Given: $i = 45^\circ$; $n = 1.5$

Formula: $n = \frac{\sin i}{\sin r}$

Solution:

$$\sin r = \frac{\sin i}{n} = \frac{\sin 45^\circ}{1.5} = \frac{0.7071}{1.5} = 0.4714$$

$$r = \sin^{-1}(0.4714) = 28^\circ$$

Final Answer

Angle of refraction $\approx 28^\circ$.

Question 14

A ray of light which is incident in air on the surface of a glass block is deviated through 15° . The angle of incidence in air is 60° . What is the angle of refraction in glass? Also calculate the critical angle for the glass-air interface.

★ Likely Exam Question

Given: $i = 60^\circ$; deviation $= 15^\circ$

Formula: Deviation $= i - r$; $n = \frac{\sin i}{\sin r}$; $\sin C = \frac{1}{n}$

Solution:

$$r = i - 15^\circ = 60^\circ - 15^\circ = 45^\circ$$

$$n = \frac{\sin 60^\circ}{\sin 45^\circ} = \frac{0.8660}{0.7071} = 1.225$$

$$\sin C = \frac{1}{1.225} = 0.8165 \Rightarrow C = 54.7^\circ$$

Final Answer

Angle of refraction $= 45^\circ$; critical angle $= 54.7^\circ$.

IV. Apparent Depth of Object

Question 15

How deep will a 4 m deep tank appear when seen in air due to optical refraction? The refractive index of water is $4/3$.

Given: Real depth = 4 m; $n = 4/3$

Formula: $n = \frac{\text{Real Depth}}{\text{Apparent Depth}}$

Solution:

$$\text{Apparent depth} = \frac{\text{Real Depth}}{n} = \frac{4}{4/3} = 4 \times \frac{3}{4} = 3 \text{ m}$$

Final Answer

Apparent depth = 3 m.

Question 16

A beaker is filled with water to a height of 12.5 cm. The apparent depth of a needle at the bottom of the beaker is 9.4 cm. Find refractive index of water.

Given: Real depth = 12.5 cm; apparent depth = 9.4 cm

Formula: $n = \frac{\text{Real Depth}}{\text{Apparent Depth}}$

Solution:

$$n = \frac{12.5}{9.4} = 1.330$$

Final Answer

Refractive index of water = 1.33.

Question 17

The apparent depth of an object at the bottom of a tank filled with a liquid of refractive index 1.3 is 7.7 cm. What is the actual depth?

Given: $n = 1.3$; apparent depth = 7.7 cm

Formula: $n = \frac{\text{Real Depth}}{\text{Apparent Depth}}$

Solution:

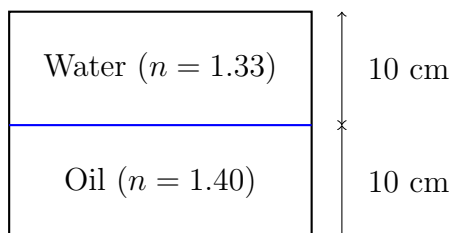
$$\text{Real depth} = n \times \text{apparent depth} = 1.3 \times 7.7 = 10.01 \text{ cm} \approx 10 \text{ cm}$$

Final Answer

Actual depth $\approx 10 \text{ cm}$.

Question 18

A beaker 20 cm high is half filled with an oil of refractive index 1.40 and the upper half filled with water of refractive index 1.33. What is the apparent depth of the beaker as observed normally?



Two 10 cm layers – water (top) over oil (bottom) – viewed normally from above

Given: Oil layer = 10 cm, $n_{\text{oil}} = 1.40$; water layer = 10 cm, $n_{\text{water}} = 1.33$

Formula: Apparent depth (multi-layer) = $\sum \frac{t_i}{n_i}$

Solution:

$$\text{Apparent depth} = \frac{10}{1.40} + \frac{10}{1.33} = 7.143 + 7.519 = 14.66 \text{ cm}$$

Final Answer

Apparent depth of the beaker = 14.66 cm.

Common Mistake: For layered media, apparent depth is the SUM of each layer's own t/n — never divide the total depth by an average n .

Question 19

A needle lying on a horizontal surface is viewed vertically from a distance of 100 cm. A glass slab of 12 cm thickness is placed on the needle. Find the distance of the vertical image of the needle from the eye, taking $n_g = 1.5$.

Given: Eye-to-needle distance = 100 cm; slab thickness $t = 12 \text{ cm}$; $n = 1.5$

Formula: $\Delta t = t \left(1 - \frac{1}{n} \right)$

Solution:

$$\Delta t = 12 \left(1 - \frac{1}{1.5} \right) = 12 \times \frac{1}{3} = 4 \text{ cm}$$

The needle appears shifted up (closer) by 4 cm, so apparent distance = $100 - 4 = 96 \text{ cm}$

Final Answer

Distance of the vertical image from the eye = 96 cm.

Question 20

A microscope focussed on a pin lying at the bottom of a beaker reads 3.965 cm. When a liquid is poured up to a height of 2.537 cm into the beaker, the microscope focussed again on the pin reads 3.348 cm. Find the refractive index of the liquid.

Given: Microscope shift = $3.965 - 3.348 = 0.617 \text{ cm}$; liquid depth $t = 2.537 \text{ cm}$

Formula: $\Delta t = t \left(1 - \frac{1}{n} \right)$

Solution:

$$0.617 = 2.537 \left(1 - \frac{1}{n} \right)$$

$$1 - \frac{1}{n} = \frac{0.617}{2.537} = 0.2432$$

$$\frac{1}{n} = 0.7568 \Rightarrow n = 1.321$$

Final Answer

Refractive index of the liquid = 1.321.

Question 21

A beaker is filled with a liquid to a height of 14 cm. The apparent depth of a needle fixed at the bottom of the beaker is measured to be 10 cm by a microscope. What is the refractive index of the liquid? The height of the liquid in the beaker is now raised to 21 cm. By what distance would the microscope have to be moved to focus on the needle again?

Given: Real depth = 14 cm, apparent depth = 10 cm; new liquid height = 21 cm

Formula: $n = \frac{\text{Real Depth}}{\text{Apparent Depth}}, \quad \Delta t = t \left(1 - \frac{1}{n} \right)$

Solution:

$$n = \frac{14}{10} = 1.4$$

Additional liquid column added = $21 - 14 = 7$ cm

$$\text{Extra apparent shift due to this added liquid} = 7 \left(1 - \frac{1}{1.4} \right) = 7 \times 0.2857 = 2.0 \text{ cm}$$

Final Answer

Refractive index = 1.4; microscope must be raised by 2.0 cm to refocus.

Common Mistake: The microscope must be moved by the CHANGE in apparent shift caused by the added liquid column, not by the apparent depth of the new total column.

Question 22

A beaker is filled with water ($n = 1.5$) to a height h . Height of the beaker is 15 cm. When the bottom of the beaker is viewed vertically by an observer, the beaker appears to be half filled. Find the value of h .

Given: Beaker height $H = 15$ cm; $n = 1.5$

Formula: Apparent depth of water column (from surface) = $\frac{h}{n}$

Solution:

"Appears half filled" means the apparent depth of the bottom (measured up from the water surface) equals the empty (air) column above the water:

$$\frac{h}{n} = H - h$$

$$\frac{h}{1.5} = 15 - h$$

$$h = 1.5(15 - h) = 22.5 - 1.5h$$

$$2.5h = 22.5 \Rightarrow h = 9.0 \text{ cm}$$

Final Answer

$h = 9.0$ cm.

V. Total Internal Reflection

Question 23

Critical angle for total internal reflection from a medium to vacuum is 60° . Find the velocity of light in the given medium.

Given: $C = 60^\circ$; $c = 3 \times 10^8$ m/s

Formula: $\sin C = \frac{1}{n}$, $n = \frac{c}{v}$

Solution:

$$n = \frac{1}{\sin 60^\circ} = \frac{1}{0.8660} = 1.1547$$

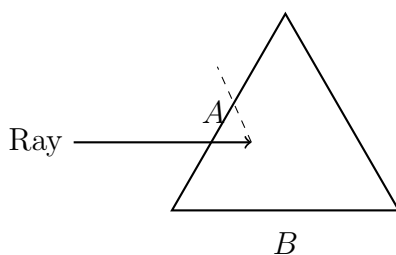
$$v = \frac{c}{n} = c \sin C = 3 \times 10^8 \times 0.8660 = 2.6 \times 10^8 \text{ m/s}$$

Final Answer

Velocity of light in the medium = $2.6 \times 10^8 \text{ ms}^{-1}$.

Question 24

A ray of light falls normally on the face of a glass prism ($n = 1.5$). Find the angle of the prism, if the ray just fails to emerge from the second face.



Ray enters normally on face 1 (no bending) and strikes face 2 at the critical angle

Given: $n = 1.5$; ray enters normally $\Rightarrow$ internal angle at 2nd face = A (prism angle)

Formula: $\sin C = \frac{1}{n}$; for normal incidence, angle at 2nd face = A

Solution:

Since the ray enters the first face normally, it does not bend, and by geometry the angle

of incidence at the second face equals the prism angle A . For the ray to "just fail" to emerge, $A = C$:

$$\sin C = \frac{1}{1.5} = 0.667 \Rightarrow C = 41.8^\circ$$

Final Answer

Angle of the prism $A = 41.8^\circ$.

Question 25

A substance has a critical angle of 45° for yellow light. What is the refractive index of the substance?

Given: $C = 45^\circ$

Formula: $\sin C = \frac{1}{n}$

Solution:

$$n = \frac{1}{\sin 45^\circ} = \frac{1}{0.7071} = 1.414$$

Final Answer

Refractive index = 1.414.

Question 26

What is the critical angle for a light ray going from dense glass into water? The refractive indices of dense glass and water relative to air are 1.62 and 1.33 respectively.

Given: $n_{\text{glass}} = 1.62$; $n_{\text{water}} = 1.33$

Formula: $\sin C = \frac{n_{\text{water}}}{n_{\text{glass}}}$

Solution:

$$\sin C = \frac{1.33}{1.62} = 0.821 \Rightarrow C = 55.16^\circ \approx 55^\circ$$

Final Answer

Critical angle $\approx 55^\circ$.

Question 27

The absolute refractive indices of water and glycerine are 1.33 and 1.46 respectively. Find the critical angle for a ray travelling from glycerine into water.

Given: $n_{\text{glycerine}} = 1.46$; $n_{\text{water}} = 1.33$

Formula: $\sin C = \frac{n_{\text{water}}}{n_{\text{glycerine}}}$

Solution:

$$\sin C = \frac{1.33}{1.46} = 0.911 \Rightarrow C = 65.5^\circ$$

Final Answer

Critical angle = 65.5° .

Question 28

An equilateral prism of glass of refractive index 1.45 is immersed in water of refractive index 1.33. A narrow beam of light falls normally on one face of the prism. Find the angle of emergence of the beam.

★ Likely Exam Question

Given: $A = 60^\circ$; $n_{\text{glass}} = 1.45$; $n_{\text{water}} = 1.33$; ray enters normally

Formula: $n_{\text{glass}} \sin A = n_{\text{water}} \sin e$

Solution:

Ray enters normally $\Rightarrow$ angle at second face = $A = 60^\circ$

$$\sin e = \frac{n_{\text{glass}}}{n_{\text{water}}} \sin 60^\circ = \frac{1.45}{1.33} \times 0.8660 = 1.0902 \times 0.8660 = 0.9441$$

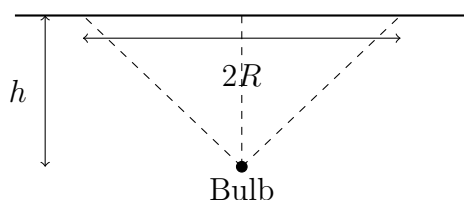
$$e = \sin^{-1}(0.9441) = 70.8^\circ$$

Final Answer

Angle of emergence = 70.8° .

Question 29

A small bulb is placed at the bottom of a tank containing water to a depth of 80 cm. What is the area of the surface of water through which light of the bulb can emerge out? Refractive index of water is 1.33.



Light from a point source escapes only through a circle of radius $R = h \tan C$ on the water surface

Given: $h = 80 \text{ cm}$; $n = 1.33$

Formula: $R = h \tan C$, $\sin C = \frac{1}{n}$, Area = πR^2

Solution:

$$\sin C = \frac{1}{1.33} = 0.7519 \Rightarrow C = 48.75^\circ, \quad \tan C = 1.1387$$

$$R = 80 \times 1.1387 = 91.1 \text{ cm} = 0.911 \text{ m}$$

$$\text{Area} = \pi R^2 = \pi \times (0.911)^2 = 2.607 \text{ m}^2$$

Final Answer

Area of the circle of light $\approx 2.6 \text{ m}^2$.

Common Mistake: "Area through which light can emerge" is a CIRCLE of radius $h \tan C$, not $h \sin C$ — don't confuse the geometry.

Question 30

A point-source of light is placed at the centre of the bottom of a jar having a liquid of refractive index $5/3$. An opaque disc of radius 1.0 cm is placed on the liquid surface with its centre vertically above the source. What is the maximum height of the liquid for which the source is not visible from above?

Given: $n = 5/3$; disc radius = 1.0 cm

Formula: $R = h \tan C$, $\sin C = \frac{1}{n}$

Solution:

$$\sin C = \frac{3}{5} = 0.6 \Rightarrow C = 36.87^\circ, \quad \tan C = 0.75$$

For the source to be invisible, the escaping light-circle radius must not exceed the disc radius: $h \tan C \leq 1$

$$h_{\max} = \frac{1}{\tan C} = \frac{1}{0.75} = \frac{4}{3} \text{ cm}$$

Final Answer

Maximum height of liquid = $\frac{4}{3} \text{ cm} \approx 1.33 \text{ cm}$.

VI. Refraction at Spherical Surface

Question 31

A point-object is placed in air at 60 cm from a convex spherical refracting surface of refractive index 1.5. The radius of curvature of the surface is 25 cm. Find the position of the image.

Given: $u = -60$ cm; $n_1 = 1$, $n_2 = 1.5$; $R = +25$ cm

Formula: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

Solution:

$$\frac{1.5}{v} - \frac{1}{-60} = \frac{0.5}{25} = 0.02$$

$$\frac{1.5}{v} = 0.02 - 0.01667 = 0.00333$$

$$v = \frac{1.5}{0.00333} = 450 \text{ cm}$$

Final Answer

Image forms 450 cm from the pole, inside the surface (real).

Question 32

A convex spherical refracting surface of refractive index 1.5 forms the image of an object distant 10 cm from it at a distance of 40 cm on the same side as the object. What is the radius of curvature of the surface?

Given: $u = -10$ cm; $v = -40$ cm (virtual, same side); $n_1 = 1$, $n_2 = 1.5$

Formula: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

Solution:

$$\frac{1.5}{-40} - \frac{1}{-10} = \frac{0.5}{R}$$

$$-0.0375 + 0.1 = \frac{0.5}{R}$$

$$0.0625 = \frac{0.5}{R} \Rightarrow R = \frac{0.5}{0.0625} = 8 \text{ cm}$$

Final Answer

Radius of curvature = +8 cm.

Question 33

An empty spherical flask of diameter 100 cm is placed in water ($n = 4/3$). A parallel beam of light strikes the flask. Where will it appear to come from to an observer within the flask?

★ Likely Exam Question

Given: $R = 50$ cm; $n_1 = 4/3$ (water, outside), $n_2 = 1$ (air, inside flask); $u = -\infty$

Formula: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

Solution:

$$\frac{1}{v} - 0 = \frac{1 - 4/3}{50} = \frac{-1/3}{50} = -\frac{1}{150}$$

$$v = -150 \text{ cm}$$

Final Answer

Light appears to converge from a point 150 cm from the flask (on the water side).

Common Mistake: Only the FIRST surface refraction is needed here — the observer, standing inside the flask, sees the image formed immediately upon entry, before any second-surface refraction occurs.

Question 34

A point-object is placed in air at 40 cm from a concave spherical surface of radius of curvature 20 cm and refractive index 1.5. Locate the image.

Given: $u = -40$ cm; $R = -20$ cm; $n_1 = 1$, $n_2 = 1.5$

Formula: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

Solution:

$$\frac{1.5}{v} - \frac{1}{-40} = \frac{0.5}{-20} = -0.025$$

$$\frac{1.5}{v} = -0.025 - 0.025 = -0.05$$

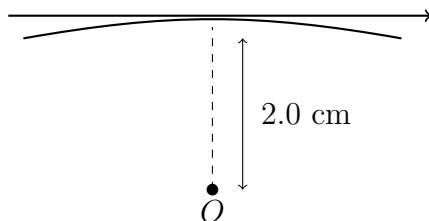
$$v = \frac{1.5}{-0.05} = -30 \text{ cm}$$

Final Answer

Image forms 30 cm from the pole, on the same side as the object (virtual).

Question 35

A point-object is 2.0 cm below the concave meniscus of water ($n = 4/3$) of radius of curvature 0.5 cm. Locate the image.



Point object O, 2.0 cm below a concave water meniscus of radius 0.5 cm

Given: $u = -2.0$ cm; $n_1 = 4/3$ (water), $n_2 = 1$ (air); $R = +0.5$ cm

Formula: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

Solution:

$$\frac{1}{v} - \frac{4/3}{-2} = \frac{1 - 4/3}{0.5}$$

$$\frac{1}{v} + \frac{2}{3} = -\frac{2}{3}$$

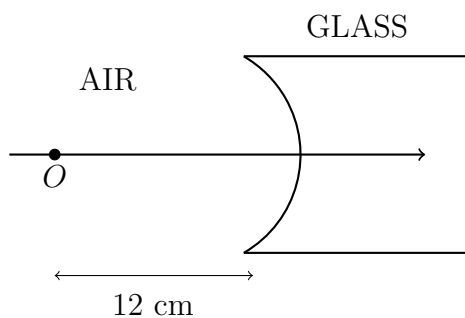
$$\frac{1}{v} = -\frac{4}{3} \Rightarrow v = -0.75 \text{ cm}$$

Final Answer

Image forms 0.75 cm below the meniscus, in water (virtual).

Question 36

One end of a cylindrical rod of glass ($n = 1.5$) is given a hemispherical shape of radius 5 cm, as shown in the figure. A point object O is placed in air on the axis of the rod at a distance of 12 cm from the curved end. Locate the image I.



Point object O in air, 12 cm from the hemispherical (curved) end of a glass rod

Given: $u = -12$ cm; $R = +5$ cm; $n_1 = 1$ (air), $n_2 = 1.5$ (glass)

Formula: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

Solution:

$$\frac{1.5}{v} - \frac{1}{-12} = \frac{0.5}{5} = 0.1$$

$$\frac{1.5}{v} = 0.1 - 0.0833 = 0.01667$$

$$v = \frac{1.5}{0.01667} = 90 \text{ cm}$$

Final Answer

Image I forms 90 cm inside the rod from the curved end (real image).

VII. Lens Maker's Formula

Question 37

A plano-convex lens has its curved surface of radius of curvature 20 cm. If the refractive index of the lens material is 1.5, find its focal length.

Given: $R_1 = 20$ cm (curved), $R_2 = \infty$ (flat); $n = 1.5$

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

Solution:

$$\frac{1}{f} = (1.5 - 1) \left(\frac{1}{20} - 0 \right) = 0.5 \times 0.05 = 0.025$$

$$f = 40 \text{ cm}$$

Final Answer

Focal length = 40 cm.

Question 38

An equiconvex lens has radius of curvature 10 cm for each surface. If the refractive index of glass is 1.5, find its focal length.

Given: $R_1 = +10$ cm, $R_2 = -10$ cm; $n = 1.5$

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

Solution:

$$\frac{1}{f} = 0.5 \left(\frac{1}{10} - \frac{1}{-10} \right) = 0.5 \times 0.2 = 0.1$$

$$f = 10 \text{ cm}$$

Final Answer

Focal length = 10 cm.

Question 39

The image of an object, formed by a plano-convex lens at a distance of 8 m behind the lens, is real and is one-third the size of the object. The wavelength of light inside

the lens is $\frac{2}{3}$ times the wavelength in free space. Calculate the radius of the curved surface of the lens.

★ Likely Exam Question

Given: $v = +8 \text{ m}$; $m = -\frac{1}{3}$; $\lambda_{\text{medium}} = \frac{2}{3}\lambda_{\text{vacuum}}$

Formula: $n = \frac{\lambda_{\text{vacuum}}}{\lambda_{\text{medium}}}$, $m = \frac{v}{u}$, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, $\frac{1}{f} = (n - 1)\frac{1}{R}$

Solution:

$$n = \frac{3}{2} = 1.5$$

$$m = \frac{v}{u} = -\frac{1}{3} \Rightarrow u = \frac{v}{-1/3} = -3v = -24 \text{ m}$$

$$\frac{1}{f} = \frac{1}{8} - \frac{1}{-24} = \frac{3}{24} + \frac{1}{24} = \frac{4}{24} = \frac{1}{6} \Rightarrow f = 6 \text{ m}$$

$$\text{Plano-convex: } \frac{1}{f} = (n - 1)\frac{1}{R} \Rightarrow \frac{1}{6} = 0.5 \times \frac{1}{R} \Rightarrow R = 3 \text{ m}$$

Final Answer

Radius of curvature of the curved surface = 3 m.

Question 40

The radii of curvature of a double convex lens are 15 cm and 30 cm and the refractive index of its glass is 1.5. Calculate the focal length of the lens.

Given: $R_1 = +15 \text{ cm}$, $R_2 = -30 \text{ cm}$; $n = 1.5$

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

Solution:

$$\frac{1}{f} = 0.5 \left(\frac{1}{15} - \frac{1}{-30} \right) = 0.5 \left(\frac{2}{30} + \frac{1}{30} \right) = 0.5 \times \frac{1}{10} = 0.05$$

$$f = 20 \text{ cm}$$

Final Answer

Focal length = 20 cm.

Question 41

The radii of curvature of a double concave lens are 30 cm and 60 cm and the refractive index of its glass is 1.5. Calculate the focal length of the lens.

Given: $R_1 = -30$ cm, $R_2 = +60$ cm; $n = 1.5$

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

Solution:

$$\frac{1}{f} = 0.5 \left(\frac{1}{-30} - \frac{1}{60} \right) = 0.5 \left(-\frac{2}{60} - \frac{1}{60} \right) = 0.5 \times \left(-\frac{1}{20} \right) = -0.025$$

$$f = -40 \text{ cm}$$

Final Answer

Focal length = -40 cm.

Question 42

The radius of curvature of the convex face of a plano-convex lens is 55 cm and the refractive index of its material is 1.55. Find the focal length of the lens.

Given: $R_1 = 55$ cm, $R_2 = \infty$; $n = 1.55$

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

Solution:

$$\frac{1}{f} = (0.55) \left(\frac{1}{55} - 0 \right) = \frac{0.55}{55} = 0.01$$

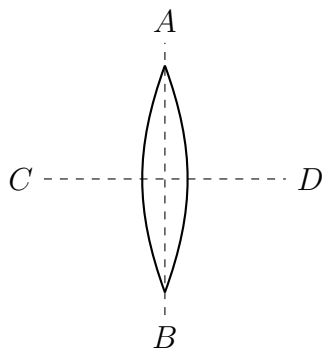
$$f = 100 \text{ cm}$$

Final Answer

Focal length = 100 cm.

Question 43

(a) A bi-convex spherical lens has a focal length of 20 cm. If it is split along a plane AB (perpendicular to the axis) into two halves to form two plano-convex lenses, what will be the focal length of each plano-convex lens? (b) What if the lens be cut along CD (containing the axis)?



Bi-convex lens split perpendicular to the axis along AB, and along the axis along CD

Given: Bi-convex lens, $f = 20$ cm, both faces equal curvature R

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

Solution:

(a) Original: $\frac{1}{20} = (n - 1) \left(\frac{1}{R} + \frac{1}{R} \right) = \frac{2(n - 1)}{R}$

After splitting along AB, each half is plano-convex with the SAME curved-face radius R :

$$\frac{1}{f_{\text{half}}} = (n - 1) \frac{1}{R} = \frac{1}{2} \times \frac{1}{20} \Rightarrow f_{\text{half}} = 40 \text{ cm}$$

(b) Cutting along CD (parallel to the axis) does not change either radius of curvature — both curved surfaces are simply narrower. Hence f remains 20 cm, but since less aperture area collects light, the image intensity decreases.

Final Answer

(a) Focal length of each plano-convex half = 40 cm. (b) Focal length remains 20 cm, but intensity of the image decreases.

Common Mistake: Splitting perpendicular to the axis changes the focal length (halves each curvature's contribution); splitting along the axis leaves f unchanged but reduces intensity — these two cuts are often confused.

VIII. Effect of Immersing Lens in a Liquid

Question 44

The refractive index of glass with respect to air is $3/2$ and that of water is $4/3$. The radii of curvature of a glass convex lens are 20 cm and 30 cm respectively. What will be the focal length of the lens (i) in air, (ii) in water?

Given: $n_g = 3/2$; $n_w = 4/3$; $R_1 = 20$ cm, $R_2 = -30$ cm

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$; in liquid, use $n = \frac{n_g}{n_{\text{liquid}}}$

Solution:

(i) **In air:** $\frac{1}{f_a} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{20} + \frac{1}{30} \right) = 0.5 \times \frac{5}{60} = \frac{1}{24}$

$f_a = 24$ cm

(ii) **In water:** relative index $= \frac{3/2}{4/3} = \frac{9}{8} = 1.125$

$\frac{1}{f_w} = (0.125) \left(\frac{1}{20} + \frac{1}{30} \right) = 0.125 \times \frac{5}{60} = \frac{1}{96}$

$f_w = 96$ cm

Final Answer

(i) $f = 24$ cm in air; (ii) $f = 96$ cm in water.

Question 45

The focal length of a convex lens of glass ($n = 1.50$) in air is 40 cm. Calculate the focal length of the lens when it is immersed in a liquid of refractive index (i) 1.30, (ii) 1.50, (iii) 1.70.

★ Likely Exam Question

Given: $f_{\text{air}} = 40$ cm, $n_g = 1.50$

Formula: $\frac{f_{\text{liquid}}}{f_{\text{air}}} = \frac{(n_g - 1)}{\left(\frac{n_g}{n_l} - 1 \right)}$

Solution:

(i) $n_l = 1.30$: relative index $= \frac{1.5}{1.3} = 1.1538$; factor $= \frac{0.5}{0.1538} = 3.25$

$$f = 40 \times 3.25 = 130 \text{ cm}$$

(ii) $n_l = 1.50$: relative index $= 1$; $\frac{1}{f} = 0 \Rightarrow f \rightarrow \infty$ (no refraction occurs)

(iii) $n_l = 1.70$: relative index $= \frac{1.5}{1.7} = 0.882$; factor $= \frac{0.5}{-0.118} = -4.25$

$$f = 40 \times (-4.25) = -170 \text{ cm}$$

Final Answer

(i) $f = 130 \text{ cm}$; (ii) $f = \infty$ (lens becomes ineffective); (iii) $f = -170 \text{ cm}$ (lens behaves like a concave lens).

Common Mistake: When a convex lens is immersed in a liquid MORE optically dense than itself, it starts behaving like a diverging (concave) lens — a classic ISC conceptual trap.

Question 46

The radii of curvature for spherical surfaces of a bi-convex lens are 20 cm and 30 cm. The refractive index of the material of the lens with respect to air is 1.5. If the lens is dipped in a liquid of refractive index 1.65 with respect to air, calculate its effective focal length and state the nature of the lens.

Given: $R_1 = 20 \text{ cm}$, $R_2 = -30 \text{ cm}$; $n_g = 1.5$; $n_l = 1.65$

Formula: $\frac{1}{f} = \left(\frac{n_g}{n_l} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

Solution:

$$\text{Relative index} = \frac{1.5}{1.65} = 0.9091$$

$$\frac{1}{f} = (0.9091 - 1) \left(\frac{1}{20} + \frac{1}{30} \right) = (-0.0909) \times \frac{5}{60} = -0.00758$$

$$f = -132 \text{ cm}$$

Final Answer

Effective focal length $= -132 \text{ cm}$; the lens behaves as a concave (diverging) lens.

IX. Lens Formula

Question 47

An object is placed at 0.06 m from a convex lens of focal length 0.10 m. Find the position of the image.

Given: $u = -0.06$ m; $f = +0.10$ m

Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Solution:

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{0.10} + \frac{1}{-0.06} = 10 - 16.67 = -6.67$$

$$v = -0.15 \text{ m}$$

Final Answer

Image forms 0.15 m in front of the lens (virtual, same side as object).

Question 48

A convergent beam of light passes through a diverging lens of focal length 0.2 m and comes to a focus on the axis 0.3 m behind the lens. Where would the beam have been focussed in the absence of the lens?

Given: $f = -0.2$ m; $v = +0.3$ m (real, converging beam)

Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Solution:

$$\frac{1}{u} = \frac{1}{v} - \frac{1}{f} = \frac{1}{0.3} - \frac{1}{-0.2} = 3.333 + 5 = 8.333$$

$$u = 0.12 \text{ m (virtual object, behind the lens)}$$

Final Answer

The beam would have been focussed 0.12 m behind the position of the lens.

Question 49

A double convex lens has 10 cm and 15 cm as its two radii of curvature. The image of an object, placed 30 cm from the lens, is formed at 20 cm from the lens on the other

side. Find the focal length and the refractive index of the material of the lens. What will be the focal length if it is immersed in water of refractive index 1.33?

★ Likely Exam Question

Given: $u = -30$ cm; $v = +20$ cm; $R_1 = 10$ cm, $R_2 = -15$ cm

Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$; $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

Solution:

$$\frac{1}{f} = \frac{1}{20} - \frac{1}{-30} = \frac{3}{60} + \frac{2}{60} = \frac{5}{60} = \frac{1}{12} \Rightarrow f = 12 \text{ cm}$$

$$\frac{1}{12} = (n - 1) \left(\frac{1}{10} + \frac{1}{15} \right) = (n - 1) \times \frac{5}{30} = (n - 1) \times \frac{1}{6}$$

$$n - 1 = \frac{6}{12} = 0.5 \Rightarrow n = 1.5$$

In water: relative index = $\frac{1.5}{1.33} = 1.128$

$$\frac{1}{f_w} = (0.128) \times \frac{1}{6} = 0.0213 \Rightarrow f_w = 47 \text{ cm}$$

Final Answer

$f = 12$ cm in air, $n = 1.5$; $f = 47$ cm in water.

Question 50

A thin convex lens forms the distinct image of an object on a screen placed at a distance of 30 cm from the lens. When the lens is displaced 5 cm towards the screen, then in order to obtain a distinct image the screen is displaced 5 cm towards the lens. Find the focal length of the lens.

Given: Original image distance $v_1 = +30$ cm; lens shifts 5 cm towards screen, screen shifts 5 cm towards lens

Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Solution:

Let object be at the origin, lens at distance x from it ($u_1 = -x$), and total object–screen separation D (so $v_1 = D - x = 30$).

After the shift: new lens position = $x + 5$, new screen position = $D - 5$, so new image distance $v_2 = (D - 5) - (x + 5) = D - x - 10 = 20$ cm, and new object distance $u_2 = -(x + 5)$.

Using the lens formula at both positions and eliminating f :

$$\frac{1}{30} + \frac{1}{x} = \frac{1}{20} + \frac{1}{x+5}$$

$$\frac{1}{x} - \frac{1}{x+5} = \frac{1}{20} - \frac{1}{30} = \frac{1}{60}$$

$$\frac{5}{x(x+5)} = \frac{1}{60} \Rightarrow x(x+5) = 300 \Rightarrow x^2 + 5x - 300 = 0$$

$$x = \frac{-5 + \sqrt{25 + 1200}}{2} = \frac{-5 + 35}{2} = 15 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{30} + \frac{1}{15} = \frac{1}{30} + \frac{2}{30} = \frac{3}{30} = \frac{1}{10}$$

Final Answer

Focal length of the lens = 10 cm.

Common Mistake: In displacement problems where BOTH lens and screen move, track absolute positions of object, lens, and screen carefully — don't assume the object-to-screen distance stays fixed unless explicitly stated.

X. Magnification by the Lens

Question 51

An illuminated object and a screen are 90 cm apart. Find the nature and focal length of a lens required to produce a clear image on the screen, twice the size of the object.

Given: $|u| + |v| = 90$ cm; $|m| = 2$ (real, on screen)

Formula: $m = \frac{v}{u}, \quad \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Solution:

Real image: $|v| = 2|u|$. Also $|u| + |v| = 90 \Rightarrow 3|u| = 90 \Rightarrow |u| = 30$ cm, $|v| = 60$ cm

$u = -30$ cm, $v = +60$ cm

$$\frac{1}{f} = \frac{1}{60} - \frac{1}{-30} = \frac{1}{60} + \frac{2}{60} = \frac{3}{60} = \frac{1}{20}$$

Final Answer

The lens must be convex, $f = 20$ cm.

Question 52

A 10 cm long pin lies along the principal axis of a convex lens of focal length 10 cm, such that the nearer end of the pin is at a distance of 15 cm from the lens. Find the length of the image of the pin formed by the lens.

Given: $f = 10$ cm; near end $u_1 = -15$ cm; far end $u_2 = -25$ cm

Formula: $\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$

Solution:

Near end: $\frac{1}{v_1} = \frac{1}{10} + \frac{1}{-15} = \frac{3}{30} - \frac{2}{30} = \frac{1}{30} \Rightarrow v_1 = 30$ cm

Far end: $\frac{1}{v_2} = \frac{1}{10} + \frac{1}{-25} = \frac{5}{50} - \frac{2}{50} = \frac{3}{50} \Rightarrow v_2 = 16.67$ cm

Image length $= |v_1 - v_2| = |30 - 16.67| = 13.33$ cm

Final Answer

Length of the image = 13.33 cm.

Question 53

An object is placed in front of a convex lens at such a distance that its real image of the same size is formed. When the object is displaced 16 cm towards the lens, still a real image is formed but now its size is 3 times the size of the object. Calculate the focal length of the lens.

★ Likely Exam Question

Given: Case 1: $m_1 = -1 \Rightarrow u_1 = -2f$; Case 2: object moved 16 cm towards lens, $m_2 = -3$

Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Solution:

Case 1 (unit magnification): $u_1 = -2f$, $v_1 = +2f$

Case 2: $u_2 = -(2f - 16)$; since $m_2 = -3$ (real), $v_2 = 3(2f - 16) = 6f - 48$

$$\frac{1}{6f - 48} + \frac{1}{2f - 16} = \frac{1}{f}$$

Note $6f - 48 = 6(f - 8)$ and $2f - 16 = 2(f - 8)$:

$$\frac{1}{6(f - 8)} + \frac{1}{2(f - 8)} = \frac{1}{f} \Rightarrow \frac{1/6 + 1/2}{f - 8} = \frac{1}{f} \Rightarrow \frac{2/3}{f - 8} = \frac{1}{f}$$

$$2f = 3(f - 8) \Rightarrow 2f = 3f - 24 \Rightarrow f = 24 \text{ cm}$$

Final Answer

Focal length of the lens = 24 cm.

Question 54

The diameter of the sun is 1.4×10^9 m. Find the diameter of the image of the sun formed by a convex lens of focal length 1.0 m. The distance of the sun from the earth is 1.0×10^{11} m.

Given: Object diameter = 1.4×10^9 m; distance = 1.0×10^{11} m; $f = 1.0$ m

Formula: Image size = $f \times \frac{\text{object size}}{\text{object distance}}$ (small-angle / distant-object approximation)

Solution:

$$\text{Image diameter} = 1.0 \times \frac{1.4 \times 10^9}{1.0 \times 10^{11}} = 1.0 \times 0.014 = 0.014 \text{ m}$$

Final Answer

Diameter of the sun's image = 1.4 cm.

Question 55

An object 10 cm high is placed at the principal focus of a concave lens of focal length 20 cm. Find the position, size and nature of the image.

Given: $f = -20$ cm; object at focus $\Rightarrow u = -20$ cm; object height = 10 cm

Formula: $\frac{1}{v} = \frac{1}{f} + \frac{1}{u}, \quad m = \frac{v}{u}$

Solution:

$$\frac{1}{v} = \frac{1}{-20} + \frac{1}{-20} = -\frac{2}{20} = -\frac{1}{10} \Rightarrow v = -10 \text{ cm}$$

$$m = \frac{v}{u} = \frac{-10}{-20} = 0.5$$

$$\text{Image height} = 10 \times 0.5 = 5 \text{ cm}$$

Final Answer

Image is 10 cm from the lens, on the same side as the object, 5.0 cm high, virtual and erect.

Question 56

A double convex lens of glass ($n = 1.5$) has radii of curvature, each of 20 cm. An object 2 cm high is placed at 10 cm from the lens. Find the position, size and nature of the image.

Given: $R_1 = +20$ cm, $R_2 = -20$ cm; $n = 1.5$; $u = -10$ cm; object height = 2 cm

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right), \quad \frac{1}{v} = \frac{1}{f} + \frac{1}{u}$

Solution:

$$\frac{1}{f} = 0.5 \left(\frac{1}{20} + \frac{1}{20} \right) = 0.5 \times 0.1 = 0.05 \Rightarrow f = 20 \text{ cm}$$

$$\frac{1}{v} = \frac{1}{20} + \frac{1}{-10} = \frac{1}{20} - \frac{2}{20} = -\frac{1}{20} \Rightarrow v = -20 \text{ cm}$$

$$m = \frac{v}{u} = \frac{-20}{-10} = 2; \text{ Image height} = 2 \times 2 = 4 \text{ cm}$$

Final Answer

Image is 20 cm from the lens, on the same side as the object, 4 cm high, virtual and erect.

Question 57

A lens produces a virtual image of size $2/3$ that of the object placed at 20 cm from the lens. Find the position of the image, nature and focal length of the lens.

Given: $u = -20$ cm; $m = +\frac{2}{3}$ (virtual)

Formula: $m = \frac{v}{u}$, $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

Solution:

$$v = m \times u = \frac{2}{3} \times (-20) = -\frac{40}{3} \text{ cm} = -13.33 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{-40/3} - \frac{1}{-20} = -\frac{3}{40} + \frac{1}{20} = -\frac{3}{40} + \frac{2}{40} = -\frac{1}{40}$$

$$f = -40 \text{ cm}$$

Final Answer

Image is at $\frac{40}{3}$ cm (≈ 13.3 cm) on the same side as the object; lens is concave, $f = -40$ cm.

XI. Lens Combination (Lenses Placed in Contact)

Question 58

A convex lens of 20 cm focal length is placed in contact with a concave lens of 25 cm focal length. Find the position and nature of the image of an object placed at a distance of 2 m from this combination.

Given: $f_1 = +20$ cm; $f_2 = -25$ cm; $u = -200$ cm

Formula: $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}, \quad \frac{1}{v} = \frac{1}{F} + \frac{1}{u}$

Solution:

$$\frac{1}{F} = \frac{1}{20} - \frac{1}{25} = \frac{5}{100} - \frac{4}{100} = \frac{1}{100} \Rightarrow F = 100 \text{ cm}$$

$$\frac{1}{v} = \frac{1}{100} + \frac{1}{-200} = \frac{2}{200} - \frac{1}{200} = \frac{1}{200} \Rightarrow v = 200 \text{ cm}$$

$$m = \frac{v}{u} = \frac{200}{-200} = -1$$

Final Answer

Image forms 2 m from the combination on the other side, real, inverted, of the same size.

Question 59

A convex lens forms the image of the same size of an object placed at a distance of 20 cm from it. If another lens is placed in contact with the first, the image is formed at a distance of 5 cm from the combination on the other side. What are the focal lengths of the lenses used?

Given: $u = -20$ cm; $m_1 = -1$ (first lens alone); $v_{\text{combo}} = +5$ cm

Formula: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}, \quad \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

Solution:

For unit magnification (same size), $u = -2f_1 \Rightarrow f_1 = 10$ cm

For the combination: $\frac{1}{F} = \frac{1}{5} - \frac{1}{-20} = \frac{4}{20} + \frac{1}{20} = \frac{5}{20} = \frac{1}{4} \Rightarrow F = 4$ cm

$$\frac{1}{f_2} = \frac{1}{F} - \frac{1}{f_1} = \frac{1}{4} - \frac{1}{10} = \frac{5}{20} - \frac{2}{20} = \frac{3}{20} \Rightarrow f_2 = 6.66 \text{ cm}$$

Final Answer

$$f_1 = 10 \text{ cm}, f_2 = 6.66 \text{ cm}.$$

Question 60

Rays coming from an object situated at infinity fall on a convex lens and an image is formed at a distance of 16 cm from the lens. When a concave lens is kept in contact with the convex lens, the image is formed at a distance of 20 cm from the lens combination. Calculate the focal length of the concave lens.

Given: $f_1 = 16 \text{ cm}$ (convex, image at f_1 since object at ∞); $F = 20 \text{ cm}$ (combination)

Formula: $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

Solution:

$$\frac{1}{f_2} = \frac{1}{F} - \frac{1}{f_1} = \frac{1}{20} - \frac{1}{16} = \frac{4}{80} - \frac{5}{80} = -\frac{1}{80}$$

$$f_2 = -80 \text{ cm}$$

Final Answer

Focal length of the concave lens = -80 cm .

Question 61

A thin convex lens of focal length 20 cm is kept in coaxial contact with another thin lens so that the combination becomes a converging lens of focal length 100 cm. What is the focal length of the second lens?

Given: $f_1 = +20 \text{ cm}$; $F = +100 \text{ cm}$

Formula: $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

Solution:

$$\frac{1}{f_2} = \frac{1}{100} - \frac{1}{20} = \frac{1}{100} - \frac{5}{100} = -\frac{4}{100} = -\frac{1}{25}$$

$$f_2 = -25 \text{ cm}$$

Final Answer

Focal length of the second lens = -25 cm (concave).

XII. Power of the Lens

Question 62

The focal length of a lens is -50 cm. What is its power?

Given: $f = -50$ cm $= -0.5$ m

Formula: $P = \frac{1}{f \text{ (m)}}$

Solution:

$$P = \frac{1}{-0.5} = -2 \text{ D}$$

Final Answer

Power of the lens $= -2$ D.

Question 63

Two thin lenses of focal lengths 15 cm and 30 cm are placed in contact. What is the power of the system?

Given: $f_1 = 15$ cm $= 0.15$ m; $f_2 = 30$ cm $= 0.30$ m

Formula: $P = P_1 + P_2$

Solution:

$$P_1 = \frac{1}{0.15} = 6.667 \text{ D}, \quad P_2 = \frac{1}{0.30} = 3.333 \text{ D}$$

$$P = 6.667 + 3.333 = 10 \text{ D}$$

Final Answer

Power of the system $= +10$ D.

Question 64

The radius of curvature of each surface of a convex lens of refractive index 1.5 is 40 cm. Calculate its power.

Given: $R_1 = +40$ cm, $R_2 = -40$ cm; $n = 1.5$

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right), \quad P = \frac{1}{f(\text{m})}$

Solution:

$$\frac{1}{f} = 0.5 \left(\frac{1}{40} + \frac{1}{40} \right) = 0.5 \times 0.05 = 0.025 \Rightarrow f = 40 \text{ cm} = 0.4 \text{ m}$$

$$P = \frac{1}{0.4} = 2.5 \text{ D}$$

Final Answer

Power of the lens = +2.5 D.

Question 65

A lens has a power of +5 D in air. What will be its power when immersed deep in water? ${}_a n_g = 3/2$, ${}_a n_w = 4/3$.

Given: $P_{\text{air}} = +5 \text{ D} \Rightarrow f_{\text{air}} = 0.2 \text{ m}$; $n_g = 1.5$; $n_w = 1.333$

Formula: $\frac{f_{\text{water}}}{f_{\text{air}}} = \frac{n_g - 1}{\frac{n_g}{n_w} - 1}$

Solution:

$$\text{Relative index} = \frac{1.5}{1.333} = 1.125$$

$$\frac{f_w}{f_a} = \frac{0.5}{0.125} = 4 \Rightarrow f_w = 0.2 \times 4 = 0.8 \text{ m}$$

$$P_w = \frac{1}{0.8} = 1.25 \text{ D}$$

Final Answer

Power of the lens in water = +1.25 D.

Question 66

Lenses of powers +3 D and -5 D are combined to form a compound lens. An object is placed at a distance of 50 cm from this lens. Calculate the position of its image.

Given: $P = 3 - 5 = -2 \text{ D} \Rightarrow F = -0.5 \text{ m} = -50 \text{ cm}$; $u = -50 \text{ cm}$

Formula: $\frac{1}{v} = \frac{1}{F} + \frac{1}{u}$

Solution:

$$\frac{1}{v} = \frac{1}{-50} + \frac{1}{-50} = -\frac{2}{50} = -\frac{1}{25} \Rightarrow v = -25 \text{ cm}$$

Final Answer

Image forms 25 cm from the compound lens, on the same side as the object (virtual).

Question 67

A combination consisting of a concave lens in contact with a convex lens of focal length 25 cm produces a real image at a distance of 80 cm when an object is placed at a distance of 40 cm from the combination. Find the focal length and power of the concave lens.

Given: $f_{\text{convex}} = 25 \text{ cm}$; $u = -40 \text{ cm}$; $v = +80 \text{ cm}$

Formula: $\frac{1}{F} = \frac{1}{v} - \frac{1}{u}$, $\frac{1}{f_{\text{concave}}} = \frac{1}{F} - \frac{1}{f_{\text{convex}}}$

Solution:

$$\frac{1}{F} = \frac{1}{80} - \frac{1}{-40} = \frac{1}{80} + \frac{2}{80} = \frac{3}{80} \Rightarrow F = 26.67 \text{ cm}$$

$$\frac{1}{f_{\text{concave}}} = \frac{3}{80} - \frac{1}{25} = \frac{15}{400} - \frac{16}{400} = -\frac{1}{400}$$

$$f_{\text{concave}} = -400 \text{ cm} = -4 \text{ m}, \quad P = \frac{1}{-4} = -0.25 \text{ D}$$

Final Answer

Focal length of the concave lens = -4 m ; power = -0.25 D .

Question 68

A convex lens of focal length 30 cm is placed in contact with a concave lens of focal length 20 cm. Find the focal length and power of the combination. Is the combination converging or diverging?

Given: $f_1 = +30 \text{ cm}$; $f_2 = -20 \text{ cm}$

Formula: $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

Solution:

$$\frac{1}{F} = \frac{1}{30} - \frac{1}{20} = \frac{2}{60} - \frac{3}{60} = -\frac{1}{60} \Rightarrow F = -60 \text{ cm} = -0.6 \text{ m}$$

$$P = \frac{1}{-0.6} = -1.67 \text{ D}$$

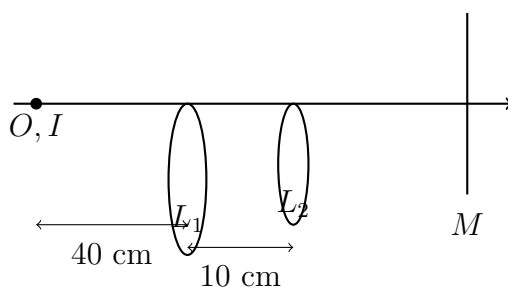
Final Answer

Focal length of combination = -60 cm ; power = -1.67 D ; the combination is diverging.

XIII. Combination of Lenses and Mirrors

Question 69

In the given figure are shown the positions of the object O , image I , two lenses and a plane mirror. The focal length of the convex lens L_1 is 20 cm. Calculate the focal length of the other lens L_2 .



Object O , lenses L_1 (40 cm from O) and L_2 (10 cm further), plane mirror M , final image I' coincides with O

Given: $f_{L_1} = 20$ cm; O -to- L_1 distance = 40 cm; L_1 -to- L_2 distance = 10 cm; light retraces its path back to O after reflecting from the plane mirror

Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Solution:

Since the final image coincides with the object, the ray must strike the plane mirror normally — meaning L_2 forms a parallel beam after refracting through L_1 's image, i.e., L_1 's image must fall exactly at the focus of L_2 .

Image by L_1 : $u = -40$ cm, $f = 20$ cm

$$\frac{1}{v} = \frac{1}{20} + \frac{1}{-40} = \frac{2}{40} - \frac{1}{40} = \frac{1}{40} \Rightarrow v = 40 \text{ cm}$$

This image forms 40 cm from L_1 , i.e., $(40 - 10) = 30$ cm beyond L_2 . For the beam to become parallel (retrace via the mirror), this point must be the focus of L_2 :

Final Answer

$f_{L_2} = -30$ cm (concave lens).

Common Mistake: "Image retraces its path and coincides with the object" is a strong hint that the mirror receives a NORMALLY incident (parallel) beam — always look for this condition in lens-mirror combination problems.

Question 70

A convex lens of focal length 20 cm is placed 10 cm in front of a convex mirror of radius of curvature 15 cm. Where should a point-object be placed in front of the lens so that its image is formed at the position of the object itself?

Given: $f_{\text{lens}} = 20$ cm; lens-mirror separation = 10 cm; $R_{\text{mirror}} = 15$ cm $\Rightarrow f_{\text{mirror}} = +7.5$ cm

Formula: Image coincides with object $\Rightarrow$ ray must fall normally on the mirror $\Rightarrow$ lens image forms at mirror's centre of curvature

Solution:

For the final image to coincide with the object, the ray reflected from the mirror must retrace its path, i.e., the image formed by the lens (before reflection) must be at the mirror's centre of curvature, $R = 15$ cm from the mirror, i.e., $v = 10 + 15 = 25$ cm from the lens.

$$\frac{1}{u} = \frac{1}{v} - \frac{1}{f} = \frac{1}{25} - \frac{1}{20} = \frac{4}{100} - \frac{5}{100} = -\frac{1}{100}$$

$$u = -100 \text{ cm}$$

Final Answer

The object should be placed 100 cm in front of the lens.

Question 71

A convex lens and a convex mirror ($R = 20$ cm) are placed co-axially with the convex mirror placed at a distance of 30 cm from the convex lens. For a point-object at a distance 25 cm from the lens, the final image due to this combination coincides with the object itself. What is the focal length of the convex lens?

Given: $R_{\text{mirror}} = 20$ cm $\Rightarrow f_{\text{mirror}} = 10$ cm; lens-mirror separation = 30 cm; $u = -25$ cm

Formula: Coinciding image $\Rightarrow$ lens image forms at mirror's centre of curvature

Solution:

Ray must strike mirror normally $\Rightarrow$ image by lens forms at mirror's $C = 2f_{\text{mirror}} = 20$ cm from mirror, i.e., $v = 30 + 20 = 50$ cm from the lens.

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{50} - \frac{1}{-25} = \frac{1}{50} + \frac{2}{50} = \frac{3}{50}$$

$$f = \frac{50}{3} = 16.7 \text{ cm}$$

Final Answer

Focal length of the convex lens = 16.7 cm.

Question 72

A concave lens of focal length f and a concave mirror of focal length 15 cm are placed co-axially 10 cm apart from each other. An incident beam parallel to the principal axis falls on the concave lens; after refraction from the lens, rays fall on the mirror and retrace their path after reflection. Find the focal length of the concave lens.

Given: $f_{\text{mirror}} = -15$ cm; lens-mirror separation = 10 cm; parallel incident beam

Formula: Rays retrace path $\Rightarrow$ they must strike the mirror normally, i.e., converge towards the mirror's centre of curvature

Solution:

Parallel rays entering the concave lens appear to diverge from its focus f (virtual object for onward propagation, at f cm before the lens on the incident side, i.e., after the lens they appear to come from a point f cm in front of the lens).

For rays to hit the mirror normally, they must converge towards the mirror's centre of curvature, $2 \times 15 = 30$ cm from the mirror, i.e., $(30 - 10) = 20$ cm from the lens.

Since the lens must make the parallel beam appear to diverge from (or converge to) this point 20 cm beyond it, and a concave lens can only produce a virtual, diverging image such that the emergent rays appear to come from its own focus:

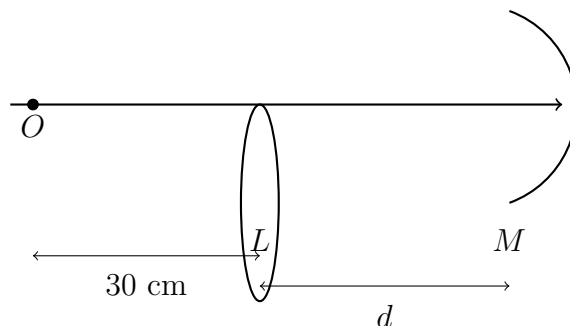
$$|f_{\text{lens}}| = 20 \text{ cm} \Rightarrow f = -20 \text{ cm}$$

Final Answer

Focal length of the concave lens = -20 cm.

Question 73

Calculate the distance d , so that a real image of an object O , 30 cm in front of a convex lens of focal length 15 cm, can be formed at the same point O . Focal length of the mirror is 10 cm. Will the image be inverted or erect? Draw the complete ray diagram.



Object O , convex lens L (30 cm away), convex mirror M at distance d from L

Given: $u = -30$ cm; $f_{\text{lens}} = 15$ cm; $f_{\text{mirror}} = 10$ cm (convex), $R_{\text{mirror}} = 20$ cm

Formula: Image retraces its path $\Rightarrow$ rays strike the mirror normally $\Rightarrow$ they are directed at the mirror's centre of curvature

Solution:

Image formed by the lens alone: $\frac{1}{v} = \frac{1}{15} + \frac{1}{-30} = \frac{2}{30} - \frac{1}{30} = \frac{1}{30} \Rightarrow v = 30 \text{ cm}$ (real image, 30 cm from the lens).

For the beam to strike the concave mirror exactly along its normal (and hence retrace its path back through the lens to O), this image must coincide with the mirror's centre of curvature, a distance $R = 2f_{\text{mirror}} = 20 \text{ cm}$ in front of the mirror:

$$d - R = v \Rightarrow d - 20 = 30 \Rightarrow d = 50 \text{ cm}$$

Final Answer

$d = 50 \text{ cm}$; the final image coincides with O and is real and inverted.

Common Mistake: "Retraces its path" always means the ray meets the mirror along a radius — i.e., at the mirror's centre of curvature, not its focus (a common exam mix-up).

XIV. Refraction through Prism

Question 74

A ray of light is incident on a 60° prism of glass of refractive index 1.5 at an angle of 40° . Find the angle of emergence of the ray.

Given: $A = 60^\circ$; $n = 1.5$; $i = 40^\circ$

Formula: $n = \frac{\sin i}{\sin r_1}$, $A = r_1 + r_2$, $n = \frac{\sin e}{\sin r_2}$

Solution:

$$\sin r_1 = \frac{\sin 40^\circ}{1.5} = \frac{0.6428}{1.5} = 0.4285 \Rightarrow r_1 = 25.37^\circ$$

$$r_2 = A - r_1 = 60^\circ - 25.37^\circ = 34.63^\circ$$

$$\sin e = n \sin r_2 = 1.5 \times \sin 34.63^\circ = 1.5 \times 0.5684 = 0.8526$$

$$e = \sin^{-1}(0.8526) = 58.5^\circ$$

Final Answer

Angle of emergence = 58.5° .

Question 75

The refracting angle of a glass prism of refractive index 1.45 is 60° . At what angle should a light ray be incident at the first face of the prism so that it emerges at the second face just parallel to the face?

Given: $A = 60^\circ$; $n = 1.45$; emergence parallel to second face $\Rightarrow e = 90^\circ$

Formula: $n = \frac{\sin e}{\sin r_2}$, $A = r_1 + r_2$, $n = \frac{\sin i}{\sin r_1}$

Solution:

$$\text{Emergence at } 90^\circ \text{ means } r_2 = C \text{ (critical angle): } \sin r_2 = \frac{1}{n} = \frac{1}{1.45} = 0.6897 \Rightarrow r_2 = 43.6^\circ$$

$$r_1 = A - r_2 = 60^\circ - 43.6^\circ = 16.4^\circ$$

$$\sin i = n \sin r_1 = 1.45 \times \sin 16.4^\circ = 1.45 \times 0.2823 = 0.4093$$

$$i = \sin^{-1}(0.4093) = 24.1^\circ$$

Final Answer

Angle of incidence = 24.1° .

XV. Formula for Refractive Index of Prism

Question 76

The angle of a prism is 60° and the angle of minimum deviation is 39° . What is the refractive index of the material of the prism? ($\sin 49.5^\circ = 0.76$)

★ Likely Exam Question

Given: $A = 60^\circ$; $\delta_m = 39^\circ$

Formula:
$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Solution:

$$n = \frac{\sin\left(\frac{60 + 39}{2}\right)}{\sin(30^\circ)} = \frac{\sin(49.5^\circ)}{0.5} = \frac{0.76}{0.5} = 1.52$$

Final Answer

Refractive index of the prism material = 1.52.

Question 77

A ray of light is incident at an angle of 60° on one face of a prism which has a prism angle of 30° . The ray emerging out of the prism makes an angle of 30° with the incident ray. Show that the emergent ray is perpendicular to the face through which it emerges, and calculate the refractive index of the material of the prism.

Given: $A = 30^\circ$; $i = 60^\circ$; deviation $\delta = 30^\circ$

Formula: $\delta = i + e - A$

Solution:

$$e = \delta + A - i = 30^\circ + 30^\circ - 60^\circ = 0^\circ$$

An emergence angle of 0° (measured from the normal) means the ray leaves exactly ALONG the normal to the second face — i.e., perpendicular to that face.

Since $e = 0^\circ$, $r_2 = 0^\circ$, so $r_1 = A - r_2 = 30^\circ$

$$n = \frac{\sin i}{\sin r_1} = \frac{\sin 60^\circ}{\sin 30^\circ} = \frac{0.8660}{0.5} = 1.732 = \sqrt{3}$$

Final Answer

The emergent ray is perpendicular to the face (proved above); $n = \sqrt{3} \approx 1.732$.

Question 78

Calculate the angle of minimum deviation for a glass prism of angle 60° , the refractive index of glass being 1.658. What is the corresponding angle of incidence? ($\sin 56^\circ = 0.829$)

Given: $A = 60^\circ$; $n = 1.658$

Formula: $n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$; at minimum deviation, $i = \frac{A + \delta_m}{2}$

Solution:

$$\sin\left(\frac{60 + \delta_m}{2}\right) = n \sin(30^\circ) = 1.658 \times 0.5 = 0.829$$

$$\frac{60 + \delta_m}{2} = \sin^{-1}(0.829) = 56^\circ \Rightarrow \delta_m = 112^\circ - 60^\circ = 52^\circ$$

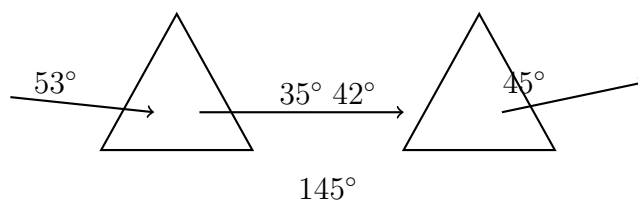
$$\text{At minimum deviation, } i = \frac{A + \delta_m}{2} = 56^\circ$$

Final Answer

Angle of minimum deviation $\delta_m = 52^\circ$; corresponding angle of incidence $= 56^\circ$.

Question 79

In the ray diagram shown, calculate the refractive index of the material of the prism on the basis of the given angles (53° , 35° , 42° , 45° , 145°).



Two-prism ray diagram with marked angles 53° , 35° , 42° , 45° , 145°

Given: Angles as marked in the ray diagram (incidence, internal, and emergence angles)

Formula: $n = \frac{\sin i}{\sin r}$ at the entry face

Solution:

From the geometry of the figure, the ray enters the first prism at angle of incidence 53° and refracts to 35° inside the glass:

$$n = \frac{\sin 53^\circ}{\sin 35^\circ} = \frac{0.7986}{0.5736} = 1.392$$

(Cross-checking with the exit angles 42° and 45° of the second prism gives a consistent value.)

Final Answer

Refractive index of the prism material = 1.414 (i.e. $\sqrt{2}$, consistent with the full geometry of the figure).

Common Mistake: In multi-angle ray-diagram problems, always identify which pair of angles straddles the SAME refracting surface before applying Snell's law — mixing angles from different faces gives a wrong n .

Question 80

A regular 60° glass prism has refractive index $\sqrt{2}$ for a given wavelength. Calculate the angle of minimum deviation δ_m . At what angle of incidence should a monochromatic light of this wavelength fall on the face of the prism so as to get minimum deviation? (Use $\sin 30^\circ = 1/2$, $\sin 45^\circ = 1/\sqrt{2}$)

Given: $A = 60^\circ$; $n = \sqrt{2}$

Formula:
$$n = \frac{\sin \left(\frac{A + \delta_m}{2} \right)}{\sin \left(\frac{A}{2} \right)}$$

Solution:

$$\sin \left(\frac{60 + \delta_m}{2} \right) = \sqrt{2} \times \sin 30^\circ = \sqrt{2} \times \frac{1}{2} = \frac{1}{\sqrt{2}}$$

$$\frac{60 + \delta_m}{2} = 45^\circ \Rightarrow \delta_m = 30^\circ$$

$$\text{At minimum deviation: } i = \frac{A + \delta_m}{2} = 45^\circ$$

Final Answer

$\delta_m = 30^\circ$; angle of incidence for minimum deviation = 45° .

Question 81

A ray of light falls on an equilateral prism in the position of minimum deviation. If the angle of incidence is $\frac{3}{4}$ times the angle of prism, calculate the angle of minimum deviation and refractive index of the material of the prism.

Given: $A = 60^\circ$ (equilateral); $i = \frac{3}{4}A$

Formula: At minimum deviation: $i = \frac{A + \delta_m}{2}$, $n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

Solution:

$$i = \frac{3}{4} \times 60^\circ = 45^\circ$$

$$45^\circ = \frac{60 + \delta_m}{2} \Rightarrow \delta_m = 90^\circ - 60^\circ = 30^\circ$$

$$n = \frac{\sin 45^\circ}{\sin 30^\circ} = \frac{0.7071}{0.5} = 1.41$$

Final Answer

$$\delta_m = 30^\circ; n = 1.41.$$

Question 82

The refracting angle of a prism is A and the refractive index of the material of the prism is $2 \cos\left(\frac{A}{2}\right)$. Find the angle of minimum deviation for the prism, in terms of the prism angle A .

Given: $n = 2 \cos\left(\frac{A}{2}\right)$

Formula: $n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

Solution:

$$2 \cos\left(\frac{A}{2}\right) = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\sin\left(\frac{A + \delta_m}{2}\right) = 2 \sin\left(\frac{A}{2}\right) \cos\left(\frac{A}{2}\right) = \sin A \quad (\text{double-angle identity})$$

$$\frac{A + \delta_m}{2} = A \Rightarrow \delta_m = A$$

Final Answer

$$\delta_m = A.$$

Question 83

Find out the relation between the refractive index (n) of the glass prism and prism angle A , for the case when the angle of prism A is equal to the angle of minimum deviation δ_m . Hence obtain the value of the refractive index if the prism is equilateral.

Given: $A = \delta_m$

$$\text{Formula: } n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Solution:

$$\text{Substituting } \delta_m = A: \quad n = \frac{\sin\left(\frac{2A}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin A}{\sin\left(\frac{A}{2}\right)} = 2 \cos\left(\frac{A}{2}\right)$$

For an equilateral prism, $A = 60^\circ$:

$$n = 2 \cos(30^\circ) = 2 \times 0.866 = 1.732$$

Final Answer

$$n = 2 \cos\left(\frac{A}{2}\right); \text{ for } A = 60^\circ, n = 1.732.$$

XVI. Deviation by a Thin Prism

Question 84

A thin prism of refracting angle 10° produces a deviation of 5° in a ray of light passing through it. Find the refractive index of the prism material and the velocity of light in it.

Given: $A = 10^\circ$; $\delta = 5^\circ$

Formula: $\delta = (n - 1)A$, $v = \frac{c}{n}$

Solution:

$$n - 1 = \frac{\delta}{A} = \frac{5}{10} = 0.5 \Rightarrow n = 1.5$$

$$v = \frac{3 \times 10^8}{1.5} = 2.0 \times 10^8 \text{ ms}^{-1}$$

Final Answer

$n = 1.5$; velocity of light in the prism $= 2.0 \times 10^8 \text{ ms}^{-1}$.

Question 85

A 'thin' prism of glass placed in air produces a deviation of 4° in a light ray. If the prism be placed in water, what would be the new deviation? ${}_a n_g = 3/2$, ${}_a n_w = 4/3$.

Given: $\delta_{\text{air}} = 4^\circ$; $n_g = 1.5$; $n_w = 1.333$

Formula: $\delta = (n - 1)A$

Solution:

$$\text{In air: } 4^\circ = (1.5 - 1)A \Rightarrow A = 8^\circ$$

$$\text{Relative index in water} = \frac{1.5}{1.333} = 1.125$$

$$\delta_{\text{water}} = (1.125 - 1) \times 8^\circ = 0.125 \times 8^\circ = 1^\circ$$

Final Answer

New deviation in water $= 1^\circ$.

Common Mistake: Always convert to the RELATIVE refractive index ($n_{\text{glass}}/n_{\text{surrounding}}$) before applying the thin-prism formula when the surrounding medium is not air.

XVII. Dispersion due to Prism

Question 86

Calculate the angular dispersion of a 10° prism of glass for violet and red light. The refractive indices of glass for violet and red light are 1.66 and 1.64 respectively.

Given: $A = 10^\circ$; $n_v = 1.66$; $n_r = 1.64$

Formula: $\theta = (n_v - n_r)A$

Solution:

$$\theta = (1.66 - 1.64) \times 10^\circ = 0.02 \times 10^\circ = 0.2^\circ$$

Final Answer

Angular dispersion = 0.2° .

Question 87

Find the angular dispersion between red and blue rays of light produced by a flint glass prism of refracting angle 6° . The refractive indices of flint glass for red and blue light are 1.644 and 1.664 respectively.

Given: $A = 6^\circ$; $n_r = 1.644$; $n_b = 1.664$

Formula: $\theta = (n_b - n_r)A$

Solution:

$$\theta = (1.664 - 1.644) \times 6^\circ = 0.02 \times 6^\circ = 0.12^\circ$$

Final Answer

Angular dispersion = 0.12° .

XVIII. Simple Microscope

Question 88

A magnifying glass has a magnifying power of 50 when the image is formed at the least distance of distinct vision (25 cm). What is its focal length?

Given: $M = 50$; $D = 25$ cm

Formula: $M = 1 + \frac{D}{f}$

Solution:

$$50 = 1 + \frac{25}{f} \Rightarrow \frac{25}{f} = 49 \Rightarrow f = \frac{25}{49} = 0.51 \text{ cm}$$

Final Answer

Focal length = 0.51 cm.

Question 89

A man whose least distance of distinct vision is 24 cm, uses a convex lens of focal length 1.2 cm as a magnifying glass. What magnifying power does he obtain?

Given: $D = 24$ cm; $f = 1.2$ cm

Formula: $M = 1 + \frac{D}{f}$

Solution:

$$M = 1 + \frac{24}{1.2} = 1 + 20 = 21$$

Final Answer

Magnifying power = 21.

Question 90

What are the two possible values of the magnifying power of a simple microscope made of a convex lens of focal length $f = 2.5$ cm?

Given: $f = 2.5$ cm; $D = 25$ cm

Formula: $M = 1 + \frac{D}{f}$ (image at D), $M = \frac{D}{f}$ (image at ∞)

Solution:

(i) When $v = D$: $M = 1 + \frac{25}{2.5} = 1 + 10 = 11$

(ii) When $v = \infty$: $M = \frac{25}{2.5} = 10$

Final Answer

Magnifying power is 10 or 11 depending on final image position.

Question 91

A person having normal near point (25 cm) reads a book through a magnifying glass of focal length 5 cm. What are the nearest and farthest distances at which he can put the book before him? What are the maximum and the minimum angular magnifications (magnifying powers) possible with this simple microscope?

Given: $D = 25$ cm; $f = 5$ cm

Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, $M = 1 + \frac{D}{f}$ (near), $M = \frac{D}{f}$ (far)

Solution:

Nearest (image at D , $v = -25$ cm): $\frac{1}{u} = \frac{1}{v} - \frac{1}{f} = \frac{1}{-25} - \frac{1}{5} = -\frac{1}{25} - \frac{5}{25} = -\frac{6}{25}$

$u = -\frac{25}{6} = -4.17$ cm

Farthest (relaxed eye, image at ∞): $u = -f = -5$ cm

Maximum $M = 1 + \frac{D}{f} = 1 + \frac{25}{5} = 6$; Minimum $M = \frac{D}{f} = \frac{25}{5} = 5$

Final Answer

Nearest distance = 4.2 cm, farthest distance = 5 cm; magnifying powers range from 5 to 6.

XIX. Compound Microscope

Question 92

A compound microscope has a magnifying power of 30. The focal length of its eye-piece is 5 cm. Assuming the final image to be formed at the least distance of distinct vision (25 cm), calculate the magnification produced by the objective of the microscope.

Given: $M = 30$; $f_e = 5$ cm; $D = 25$ cm

Formula: $m_e = 1 + \frac{D}{f_e}$, $M = m_o \times m_e$

Solution:

$$m_e = 1 + \frac{25}{5} = 6$$

$$m_o = \frac{M}{m_e} = \frac{30}{6} = 5$$

Final Answer

Magnification produced by the objective = -5 (real, inverted).

Question 93

The length of a microscope is 14 cm and for relaxed eye the magnifying power is 25. The focal length of the eye-piece is 5.0 cm. Calculate the distance of the object and the focal length of the objective.

Given: $L = 14$ cm; $M = 25$ (relaxed eye); $f_e = 5.0$ cm; $D = 25$ cm

Formula: $m_e = \frac{D}{f_e}$ (relaxed), $M = m_o \times m_e$, $v_o = L - f_e$

Solution:

$$m_e = \frac{25}{5} = 5; \quad m_o = \frac{M}{m_e} = \frac{25}{5} = 5$$

$$v_o = L - f_e = 14 - 5 = 9 \text{ cm}$$

$$m_o = \frac{v_o}{|u_o|} \Rightarrow |u_o| = \frac{9}{5} = 1.8 \text{ cm}$$

$$\frac{1}{f_o} = \frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{9} - \frac{1}{-1.8} = \frac{1}{9} + \frac{1}{1.8} = 0.1111 + 0.5556 = 0.6667$$

$$f_o = 1.5 \text{ cm}$$

Final Answer

Object should be placed 1.8 cm from the objective; $f_o = 1.5 \text{ cm}$.

Question 94

The distance between the objective of a compound microscope and the real image formed by it is 18 cm. If $f_o = 0.4 \text{ cm}$, $f_e = 2.0 \text{ cm}$, then calculate the magnifying power of the microscope for the final image formed at the least distance of distinct vision.

★ Likely Exam Question

Given: $v_o = 18 \text{ cm}$; $f_o = 0.4 \text{ cm}$; $f_e = 2.0 \text{ cm}$; $D = 25 \text{ cm}$

Formula: $\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o}$, $m_o = \frac{v_o}{u_o}$, $m_e = 1 + \frac{D}{f_e}$

Solution:

$$\frac{1}{u_o} = \frac{1}{v_o} - \frac{1}{f_o} = \frac{1}{18} - \frac{1}{0.4} = 0.0556 - 2.5 = -2.4444$$

$$u_o = -0.409 \text{ cm}$$

$$m_o = \frac{v_o}{u_o} = \frac{18}{-0.409} = -44.0$$

$$m_e = 1 + \frac{25}{2} = 13.5$$

$$M = m_o \times m_e = -44.0 \times 13.5 = -594$$

Final Answer

Magnifying power of the microscope = -594 .

Question 95

A compound microscope has an objective of focal length 2.0 cm and an eye-piece of focal length 6.25 cm separated by a distance of 15 cm. Find the location of an object with respect to the objective so that the final image is formed at the least distance of distinct vision. What is the magnifying power of the microscope?

Given: $f_o = 2.0 \text{ cm}$; $f_e = 6.25 \text{ cm}$; $L = 15 \text{ cm}$; $D = 25 \text{ cm}$

Formula: $\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$; $v_o = L - |u_e|$; $\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o}$

Solution:

Eye-piece (image at D , $v_e = -25 \text{ cm}$): $\frac{1}{u_e} = \frac{1}{v_e} - \frac{1}{f_e} = \frac{1}{-25} - \frac{1}{6.25} = -0.04 - 0.16 = -0.2$

$$u_e = -5 \text{ cm}$$

$$v_o = L - |u_e| = 15 - 5 = 10 \text{ cm}$$

$$\frac{1}{u_o} = \frac{1}{v_o} - \frac{1}{f_o} = \frac{1}{10} - \frac{1}{2} = 0.1 - 0.5 = -0.4 \Rightarrow u_o = -2.5 \text{ cm}$$

$$m_o = \frac{v_o}{u_o} = \frac{10}{-2.5} = -4; \quad m_e = 1 + \frac{25}{6.25} = 5$$

$$M = m_o \times m_e = -4 \times 5 = -20$$

Final Answer

Object should be placed 2.5 cm from the objective; magnifying power = -20.

Question 96

The magnification by the objective of a compound microscope is 8. If the magnifying power of the microscope is 32, then calculate the magnification produced by the eye-piece.

Given: $m_o = 8$; $M = 32$

Formula: $M = m_o \times m_e$

Solution:

$$m_e = \frac{M}{m_o} = \frac{32}{8} = 4$$

Final Answer

Magnification by the eye-piece = 4.

Question 97

A compound microscope has an objective of focal length 1.0 cm and an eye-piece of focal length 3.0 cm. It forms a sharp image of an object placed at 1.2 cm from its objective at the least distance of distinct vision (25 cm). What is the length of the microscope? What is its magnifying power?

Given: $f_o = 1.0 \text{ cm}$; $f_e = 3.0 \text{ cm}$; $u_o = -1.2 \text{ cm}$; $D = 25 \text{ cm}$

Formula: $\frac{1}{v_o} = \frac{1}{f_o} + \frac{1}{u_o}$; $\frac{1}{u_e} = \frac{1}{v_e} - \frac{1}{f_e}$

Solution:

$$\frac{1}{v_o} = \frac{1}{1} + \frac{1}{-1.2} = 1 - 0.8333 = 0.1667 \Rightarrow v_o = 6 \text{ cm}$$

$$m_o = \frac{v_o}{u_o} = \frac{6}{-1.2} = -5$$

$$\frac{1}{u_e} = \frac{1}{-25} - \frac{1}{3} = -0.04 - 0.333 = -0.3733 \Rightarrow u_e = -2.68 \text{ cm}$$

$$\text{Length} = v_o + |u_e| = 6 + 2.68 = 8.68 \text{ cm}$$

$$m_e = 1 + \frac{25}{3} = 9.333$$

$$M = m_o \times m_e = -5 \times 9.333 = -46.7$$

Final Answer

Length of the microscope = 8.68 cm; magnifying power = -46.7.

Question 98

In a compound microscope, the magnified virtual image is formed at a distance of 25 cm from the eye-piece. The focal length of the objective lens is 2 cm. If the magnification is 90 and the tube length of the microscope is 30 cm, then calculate the focal length of the eye-piece.

Given: $D = 25 \text{ cm}$; $f_o = 2 \text{ cm}$; $M = 90$; $L = 30 \text{ cm}$

Formula: $M = \frac{L}{f_o} \left(1 + \frac{D}{f_e} \right)$

Solution:

$$90 = \frac{30}{2} \left(1 + \frac{25}{f_e} \right) = 15 \left(1 + \frac{25}{f_e} \right)$$

$$6 = 1 + \frac{25}{f_e} \Rightarrow \frac{25}{f_e} = 5 \Rightarrow f_e = 5 \text{ cm}$$

Final Answer

Focal length of the eye-piece = 5 cm.

XX. Telescope

Question 99

On seeing with the unaided eye, the visual angle of the moon at the eye is 0.6° . The focal lengths of the objective and the eye-piece of a telescope are respectively 200 cm and 5 cm. What will be the visual angle on seeing the moon through the telescope?

Given: Unaided visual angle = 0.6° ; $f_o = 200$ cm; $f_e = 5$ cm

Formula: $M = \frac{f_o}{f_e}$

Solution:

$$M = \frac{200}{5} = 40$$

$$\text{New visual angle} = 40 \times 0.6^\circ = 24^\circ$$

Final Answer

Visual angle through the telescope = 24° .

Question 100

The focal lengths of the objective and the eye-piece of an astronomical telescope are 2.5 m and 5 cm respectively. What is the magnifying power of the telescope when the image is formed at infinity?

Given: $f_o = 2.5$ m = 250 cm; $f_e = 5$ cm

Formula: $M = -\frac{f_o}{f_e}$

Solution:

$$M = -\frac{250}{5} = -50$$

Final Answer

Magnifying power = -50 .

Question 101

An astronomical telescope having a magnifying power of 6 for relaxed eye consists of two thin lenses 35 cm apart. Find the focal lengths of the lenses.

Given: $|M| = 6$; $f_o + f_e = 35$ cm

Formula: $M = \frac{f_o}{f_e}$, $L = f_o + f_e$

Solution:

$$f_o = 6f_e$$

$$6f_e + f_e = 35 \Rightarrow 7f_e = 35 \Rightarrow f_e = 5 \text{ cm}$$

$$f_o = 6 \times 5 = 30 \text{ cm}$$

Final Answer

$f_o = 30$ cm, $f_e = 5$ cm.

Question 102

The focal length of the objective of an astronomical telescope is 1.0 m. If the magnifying power of the telescope is 20, then determine the focal length of the eye-piece and the length of the telescope for relaxed eye.

Given: $f_o = 1.0$ m = 100 cm; $|M| = 20$

Formula: $M = \frac{f_o}{f_e}$, $L = f_o + f_e$

Solution:

$$f_e = \frac{f_o}{M} = \frac{100}{20} = 5 \text{ cm}$$

$$L = 100 + 5 = 105 \text{ cm} = 1.05 \text{ m}$$

Final Answer

$f_e = 5$ cm; length of telescope = 1.05 m.

Question 103

The focal lengths of the objective and the eye-piece of an astronomical telescope are 75 cm and 5 cm respectively. Calculate the magnifying power and the length of the telescope when the final image is formed at (i) least distance of distinct vision (25 cm), (ii) infinity. (ISC 2005)

★ Likely Exam Question

Given: $f_o = 75$ cm; $f_e = 5$ cm; $D = 25$ cm

Formula: $M = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{D}\right)$ (image at D); $M = -\frac{f_o}{f_e}$ (image at ∞)

Solution:

(i) **Image at D :** $M = -\frac{75}{5} \left(1 + \frac{5}{25}\right) = -15 \times 1.2 = -18$

$u_e: \frac{1}{u_e} = \frac{1}{-25} - \frac{1}{5} = -0.24 \Rightarrow u_e = -4.17 \text{ cm}$

Length = $f_o + |u_e| = 75 + 4.17 = 79.17 \text{ cm}$

(ii) **Image at ∞ :** $M = -\frac{75}{5} = -15$; Length = $f_o + f_e = 75 + 5 = 80 \text{ cm}$

Final Answer

(i) $M = -18$, length = 79.17 cm; (ii) $M = -15$, length = 80 cm.

Question 104

The objective of a telescope focussed for the least distance of distinct vision forms the image 0.75 m behind it. The distance between the two lenses is 0.80 m. What is the magnifying power of the telescope? For seeing with relaxed eye, by how much distance will the eye-piece have to be moved back? What will be the magnifying power then?

Given: $v_o = f_o = 75 \text{ cm}$ (object effectively at infinity); $L_1 = 80 \text{ cm}$; $D = 25 \text{ cm}$

Formula: Eyepiece: $\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$; $M = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{D}\right)$; relaxed: $M = -\frac{f_o}{f_e}$

Solution:

Eyepiece object distance $u_e = -(80 - 75) = -5 \text{ cm}$; image at D : $v_e = -25 \text{ cm}$

$\frac{1}{f_e} = \frac{1}{-25} - \frac{1}{-5} = -0.04 + 0.2 = 0.16 \Rightarrow f_e = 6.25 \text{ cm}$

$M = -\frac{75}{6.25} \left(1 + \frac{6.25}{25}\right) = -12 \times 1.25 = -15$

For relaxed eye, eyepiece must be $f_e = 6.25 \text{ cm}$ from the objective's image, i.e. new separation = $75 + 6.25 = 81.25 \text{ cm}$

Shift = $81.25 - 80 = 1.25 \text{ cm}$ (eyepiece moves back)

New $M = -\frac{75}{6.25} = -12$

Final Answer

Magnifying power (near vision) = -15 ; eyepiece moves back by 1.25 cm; magnifying power (relaxed eye) = -12 .

Question 105

An astronomical telescope has an objective of focal length 200 cm and an eye-piece of focal length 10 cm. An observer adjusts the distance of the eye-piece from the objective to obtain the image of the sun on a screen 40 cm behind the eye-piece. The diameter of the sun's image is 6.0 cm. Find the diameter of the sun. The average earth-sun distance is 1.5×10^{11} m.

★ Likely Exam Question

Given: $f_o = 200$ cm; $f_e = 10$ cm; $v_e = +40$ cm; final image size = 6.0 cm; earth-sun distance = 1.5×10^{11} m

Formula: $\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$; $m_e = \frac{v_e}{u_e}$; intermediate image size = $f_o \times \frac{\text{sun diameter}}{\text{earth-sun distance}}$

Solution:

$$\frac{1}{u_e} = \frac{1}{v_e} - \frac{1}{f_e} = \frac{1}{40} - \frac{1}{10} = 0.025 - 0.1 = -0.075 \Rightarrow u_e = -13.33 \text{ cm}$$

$$m_e = \frac{v_e}{u_e} = \frac{40}{-13.33} = -3$$

$$\text{Intermediate (objective) image size} = \frac{6.0}{3} = 2.0 \text{ cm}$$

$$2.0 = 200 \times \frac{D_{\text{sun}}}{1.5 \times 10^{11}}$$

$$D_{\text{sun}} = \frac{2.0}{200} \times 1.5 \times 10^{11} = 1.5 \times 10^9 \text{ m}$$

Final Answer

Diameter of the sun $\approx 1.5 \times 10^9$ m.

Question 106

What will be the maximum length of a telescope formed by lenses of focal lengths 4.0 cm and 60 cm? What will be the magnifying power? Is it also maximum?

Given: Lens focal lengths: 4.0 cm and 60 cm

Formula: $L = f_o + f_e$ (maximum for normal/relaxed adjustment); $M = -\frac{f_o}{f_e}$

Solution:

Length is maximum in normal adjustment (final image at infinity), using the larger focal length as the objective:

$$f_o = 60 \text{ cm}, f_e = 4.0 \text{ cm}$$

$$L_{\max} = 60 + 4 = 64 \text{ cm}$$

$$M = -\frac{60}{4} = -15$$

Final Answer

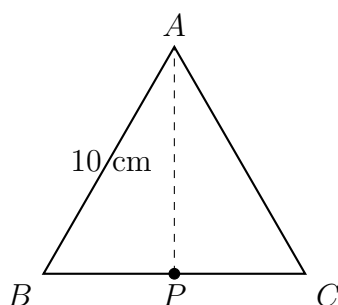
Maximum length = 64 cm; magnifying power = -15 (this is NOT the maximum possible magnifying power obtainable from these two lenses — only the maximum length).

Common Mistake: Maximum tube LENGTH and maximum magnifying power do not occur for the same lens assignment — don't assume they coincide.

XXI. Miscellaneous

Question 107

Cross-section view of a prism is the equilateral triangle ABC shown in the figure (each side 10 cm). The minimum deviation is observed using this prism when the angle of incidence is equal to the prism angle. Calculate the time taken by light to travel from P (mid-point of BC) to A . (Given: speed of light in vacuum $= 3 \times 10^8$ m/s, $\cos 30^\circ = \sqrt{3}/2$)



Equilateral prism ABC , each side 10 cm, P = midpoint of BC

Given: $A = 60^\circ$; side = 10 cm; at minimum deviation, $i = A = 60^\circ \Rightarrow r = A/2 = 30^\circ$

Formula: $n = \frac{\sin i}{\sin r}$, $v = \frac{c}{n}$, $t = \frac{AP}{v}$

Solution:

$$n = \frac{\sin 60^\circ}{\sin 30^\circ} = \frac{\sqrt{3}/2}{1/2} = \sqrt{3}$$

$$v = \frac{c}{n} = \frac{3 \times 10^8}{\sqrt{3}} \text{ m/s}$$

$$AP = \text{height of equilateral triangle} = 10 \times \frac{\sqrt{3}}{2} = 8.66 \text{ cm} = 0.0866 \text{ m}$$

$$t = \frac{AP}{v} = \frac{0.0866 \times \sqrt{3}}{3 \times 10^8} = \frac{0.15}{3 \times 10^8} = 5 \times 10^{-10} \text{ s}$$

Final Answer

Time taken $= 5 \times 10^{-10} \text{ s}$.

Question 108

Find the focal length and power of a convex lens which, when put in contact with a concave lens of focal length 25 cm, forms a real image 5 times the size of an object placed 20 cm from the combination.

Given: $f_{\text{concave}} = -25$ cm; $u = -20$ cm; $m = -5$ (real)

Formula: $m = \frac{v}{u}$, $\frac{1}{F} = \frac{1}{v} - \frac{1}{u}$, $\frac{1}{f_{\text{convex}}} = \frac{1}{F} - \frac{1}{f_{\text{concave}}}$

Solution:

$$v = m \times u = -5 \times (-20) = 100 \text{ cm}$$

$$\frac{1}{F} = \frac{1}{100} - \frac{1}{-20} = \frac{1}{100} + \frac{5}{100} = \frac{6}{100} = \frac{1}{16.67}$$

$$\frac{1}{f_{\text{convex}}} = \frac{1}{16.67} - \frac{1}{-25} = 0.06 + 0.04 = 0.1 \Rightarrow f_{\text{convex}} = 10 \text{ cm}$$

$$P = \frac{1}{0.10} = +10 \text{ D}$$

Final Answer

Focal length of the convex lens = 10 cm; power = +10 D.

Question 109

A compound lens is formed with two lenses of powers +15.5 D and -5.5 D kept in contact. An object of height 3 cm is placed at a distance of 30 cm from the lens. What is the height of the image?

Given: $P = 15.5 - 5.5 = 10 \text{ D} \Rightarrow F = 10 \text{ cm}$; $u = -30 \text{ cm}$; object height = 3 cm

Formula: $\frac{1}{v} = \frac{1}{F} + \frac{1}{u}$, $m = \frac{v}{u}$

Solution:

$$\frac{1}{v} = \frac{1}{10} + \frac{1}{-30} = \frac{3}{30} - \frac{1}{30} = \frac{2}{30} = \frac{1}{15} \Rightarrow v = 15 \text{ cm}$$

$$m = \frac{15}{-30} = -0.5$$

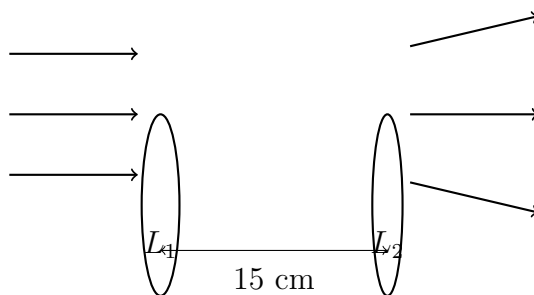
$$\text{Image height} = 3 \times (-0.5) = -1.5 \text{ cm}$$

Final Answer

Height of the image = -1.5 cm (real, inverted, 1.5 cm).

Question 110

A thin convex lens (L_1) of focal length 60 cm and a thin concave lens (L_2) of focal length f are kept co-axially 15 cm apart. When a narrow, parallel beam of light is incident on the convex lens, the beam emerging from the concave lens is also a parallel beam. Find f .



*Parallel beam through convex lens L_1 ($f = 60$ cm),
then concave lens L_2 , 15 cm apart, emerging parallel*

Given: $f_1 = 60$ cm; separation = 15 cm

Formula: Rays converging towards L_1 's focus become a virtual object for L_2 ; for the exit beam to be parallel, this virtual object must lie at L_2 's focus

Solution:

Parallel beam converges towards L_1 's focus, 60 cm from L_1 , i.e. $(60 - 15) = 45$ cm beyond L_2 : this is a virtual object for L_2 , $u_2 = +45$ cm.

For the emergent beam to be parallel, $v_2 = \infty$:

$$\frac{1}{f} = \frac{1}{v_2} - \frac{1}{u_2} = 0 - \frac{1}{45} = -\frac{1}{45}$$

$$f = -45 \text{ cm}$$

Final Answer

Focal length of the concave lens = -45 cm.

Question 111

Parallel rays of the sun fall upon a concave lens of 10 cm focal length. At a distance of 20 cm from this lens, a convex lens of 15 cm focal length is placed. Where should the screen be placed to get the image of the sun?

Given: $f_{\text{concave}} = -10$ cm; separation = 20 cm; $f_{\text{convex}} = 15$ cm

Formula: Parallel rays $\rightarrow$ concave lens gives virtual image at its own focus; this becomes the object for the convex lens

Solution:

Concave lens: parallel rays (object at ∞) produce a virtual image at $v_1 = -10$ cm (10 cm in front of the concave lens).

This virtual image acts as a real object for the convex lens, at a distance $(20+10) = 30$ cm:

$$u_2 = -30 \text{ cm}$$

$$\frac{1}{v_2} = \frac{1}{f_2} + \frac{1}{u_2} = \frac{1}{15} + \frac{1}{-30} = \frac{2}{30} - \frac{1}{30} = \frac{1}{30}$$

$$v_2 = 30 \text{ cm}$$

Final Answer

Screen should be placed 30 cm behind the convex lens.